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Hwang et al.

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(54) **METHOD AND DEVICE FOR
DETERMINING SIDELINK TRANSMISSION
RESOURCE ON BASIS OF RSRP VALUE IN
NR V2X**

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H04B 17/318 (2015.01)
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CPC **H04W 72/40** (2023.01); **H04B 17/318**
(2015.01); **H04W 72/02** (2013.01); **H04W**
72/542 (2023.01)

(58) **Field of Classification Search**
CPC ... **H04W 72/40**; **H04W 72/542**; **H04W 72/02**;
H04B 17/318
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2019/0373622 A1 12/2019 Li et al.
2020/0029340 A1* 1/2020 He **H04W 72/25**
2023/0020105 A1* 1/2023 Shin **H04W 72/542**

FOREIGN PATENT DOCUMENTS

KR 1020190000891 1/2019
KR 1020200027970 3/2020

(Continued)

OTHER PUBLICATIONS

PCT International Application No. PCT/KR2021/008708, Interna-
tional Search Report dated Oct. 18, 2021, 6 pages.

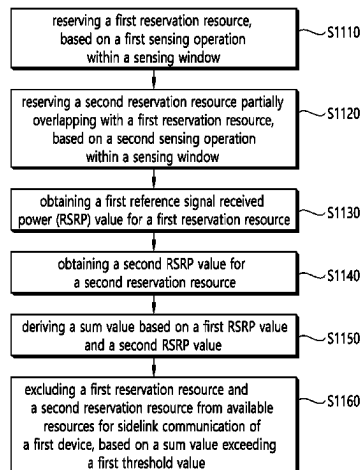
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DEGERMAN, KANG & WAIMEY

(57) **ABSTRACT**

An embodiment of the present disclosure provides a method for performing sidelink communication by a first device. The method may comprise the steps of: reserving a first reservation resource on the basis of a first sensing operation within a sensing window; on the basis of a second sensing operation within the sensing window, reserving a second reservation resource partially overlapping with the first reservation resource; obtaining a first RSRP value of the first reservation resource; obtaining a second RSRP value of the second reservation resource; deriving a sum value on the basis of the first RSRP value and the second RSRP value; and on the basis that the sum value exceeds a first threshold

(Continued)



value, excluding the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device.

12 Claims, 18 Drawing Sheets

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H04W 72/02 (2009.01)
H04W 72/542 (2023.01)

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

KR	1020200057011	5/2020
WO	2017150959	9/2017

* cited by examiner

FIG. 1

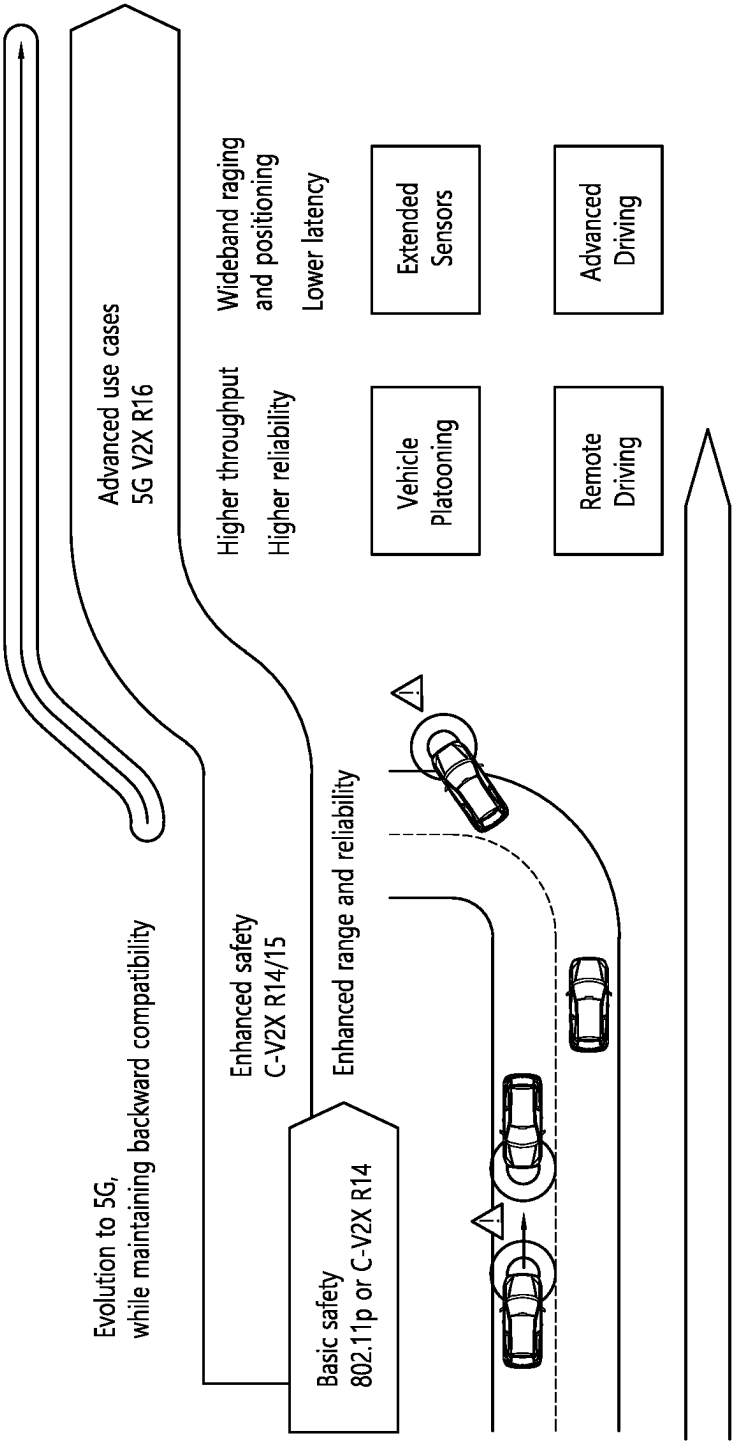


FIG. 2

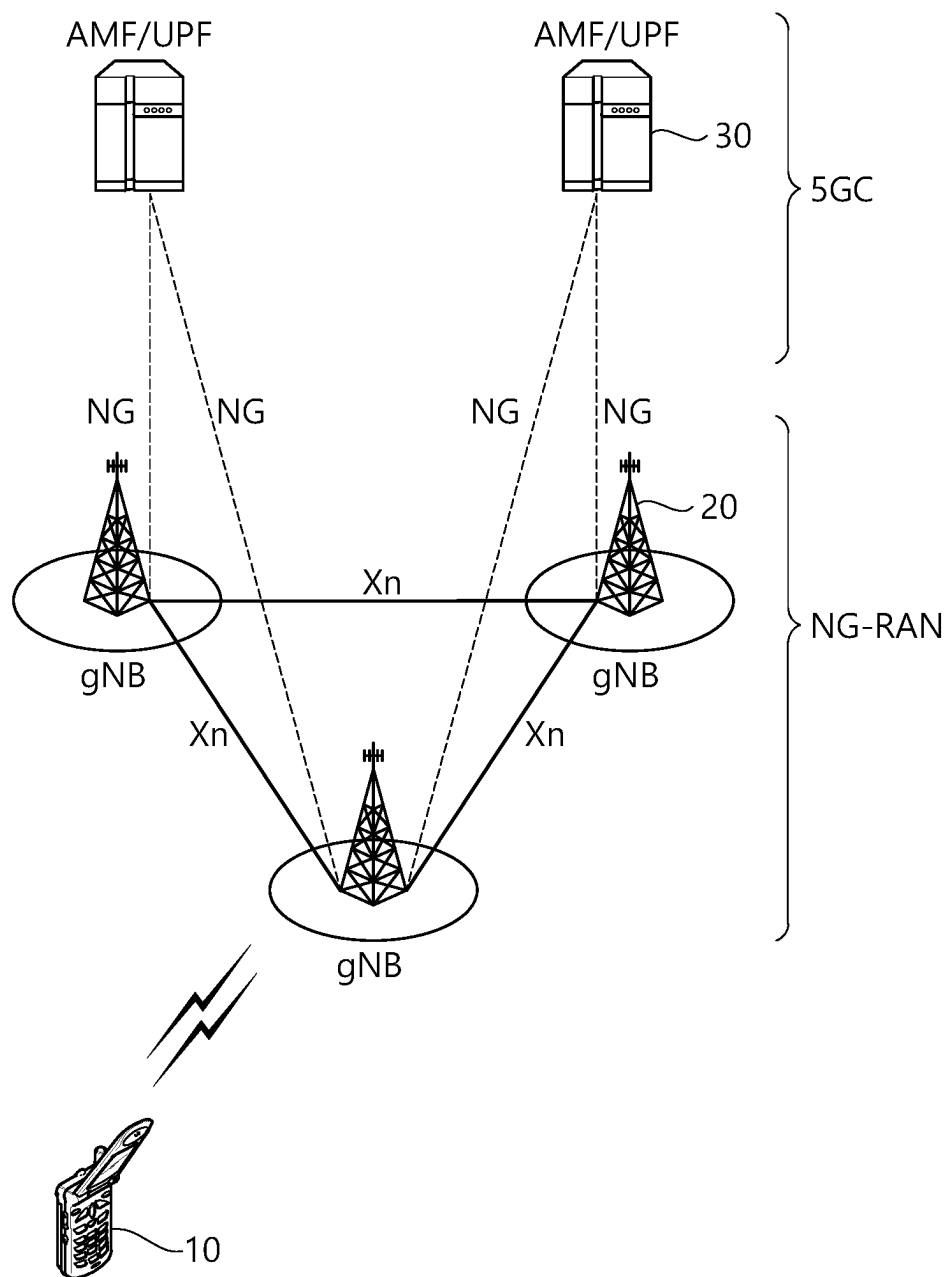


FIG. 3

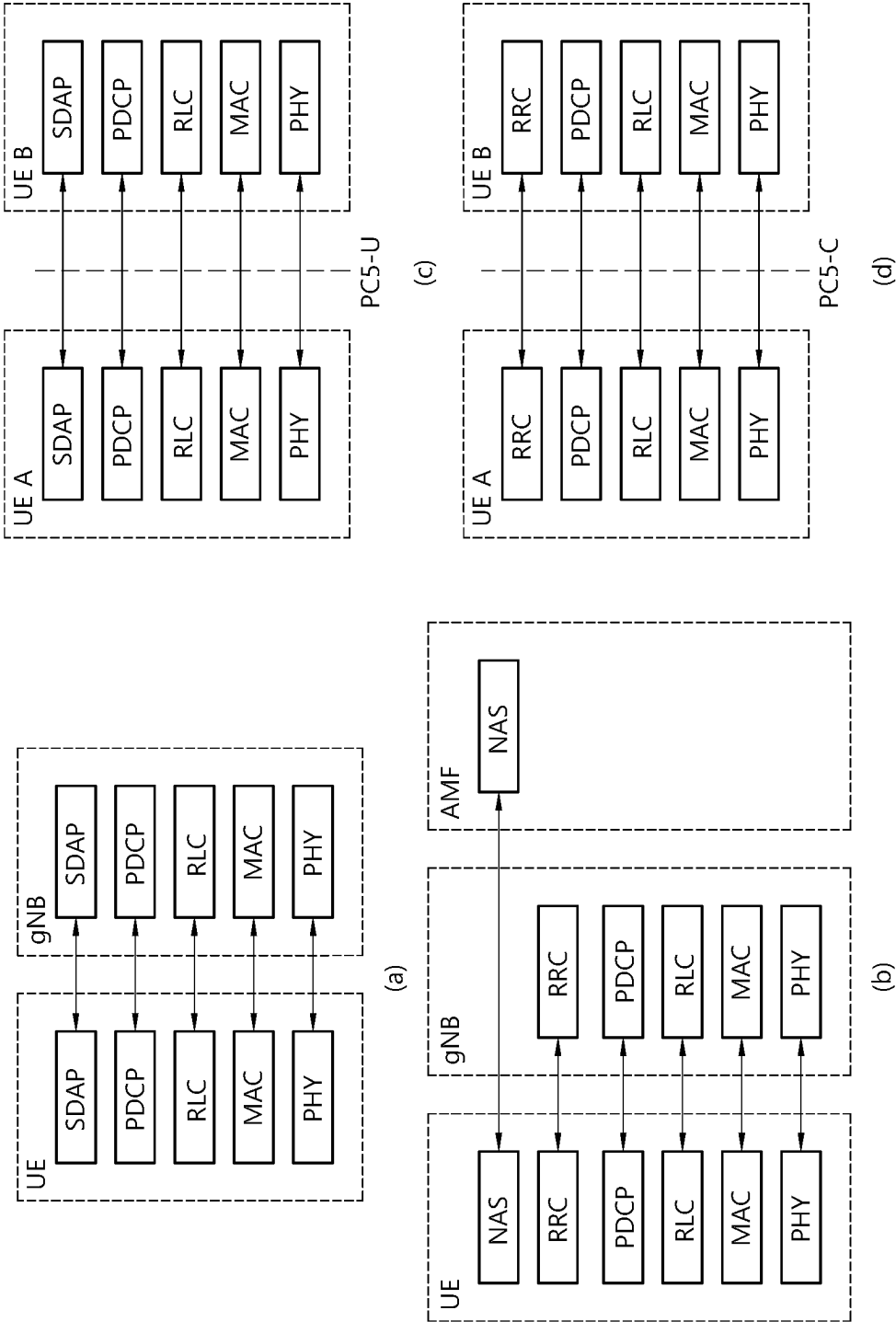


FIG. 4

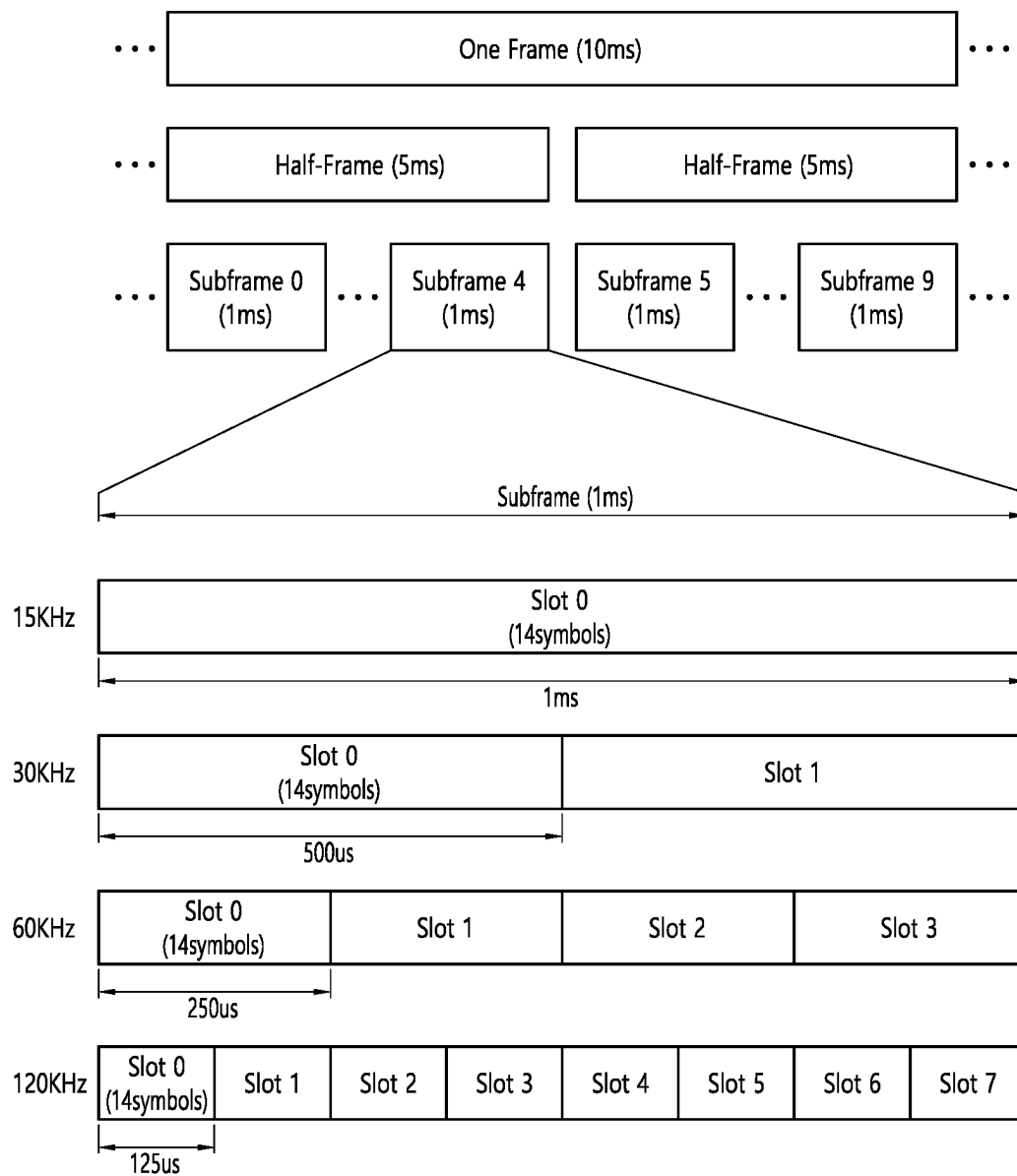


FIG. 5

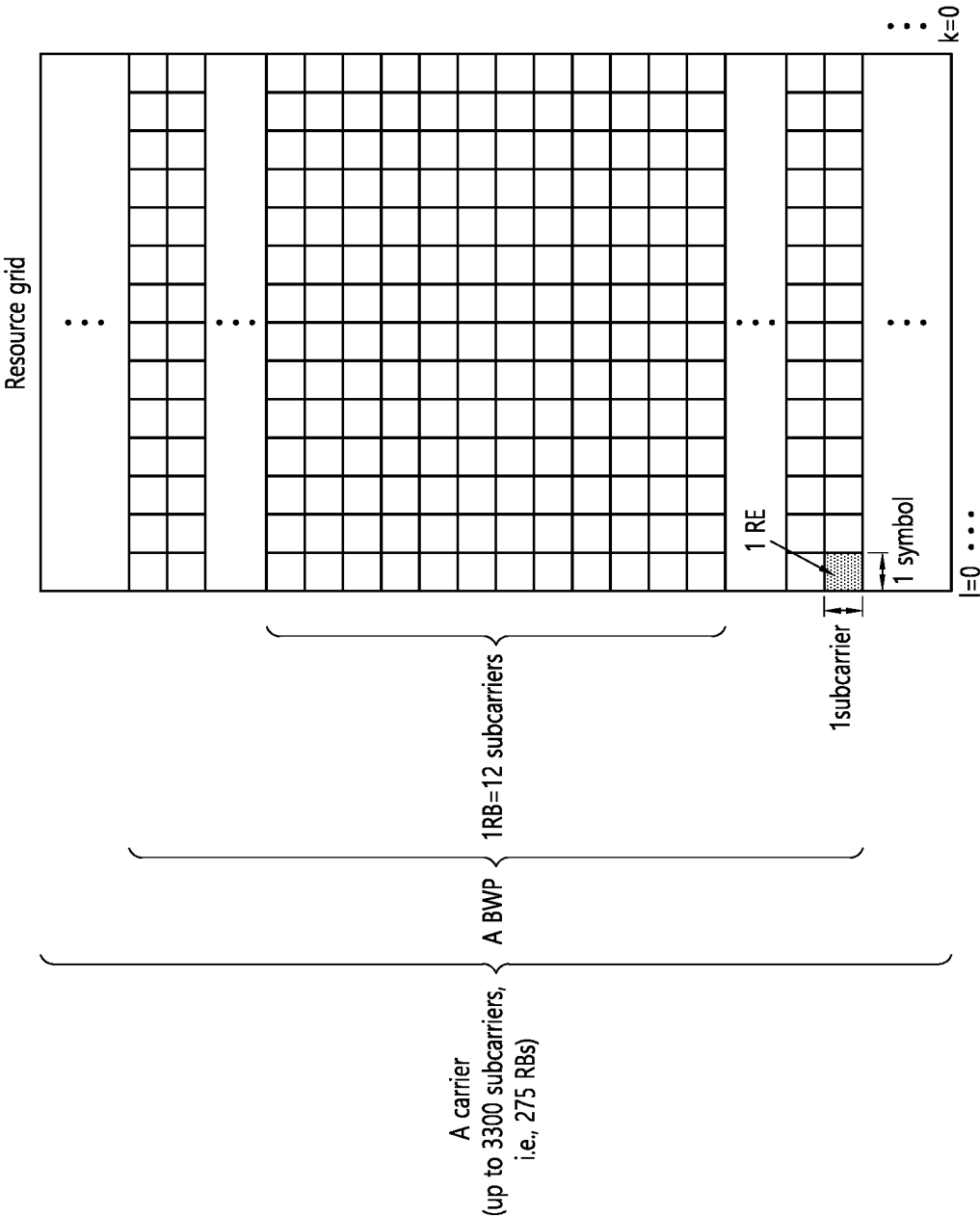


FIG. 6

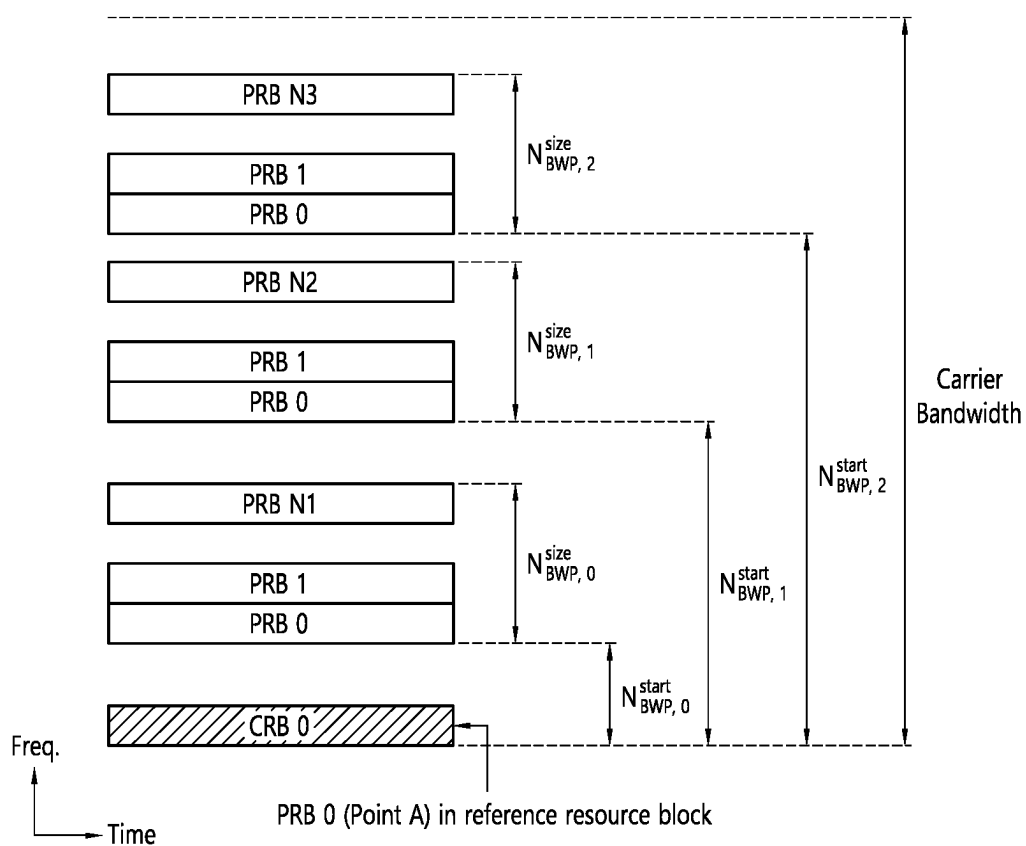


FIG. 7

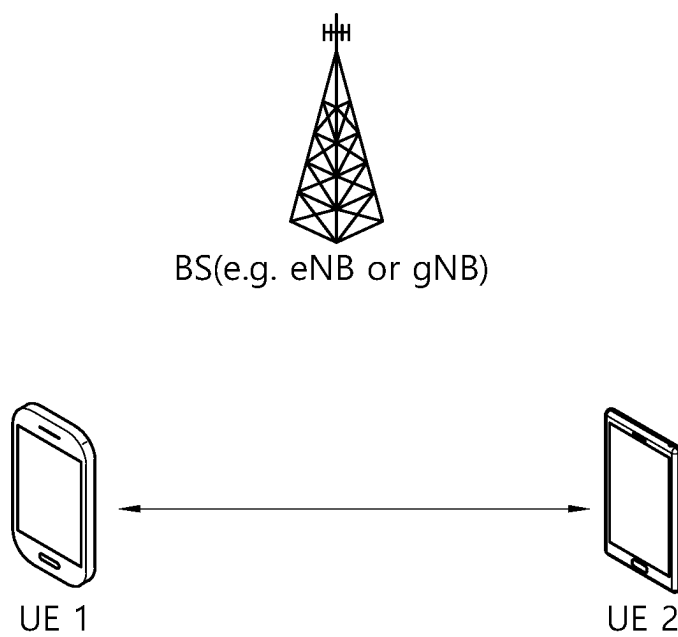


FIG. 8

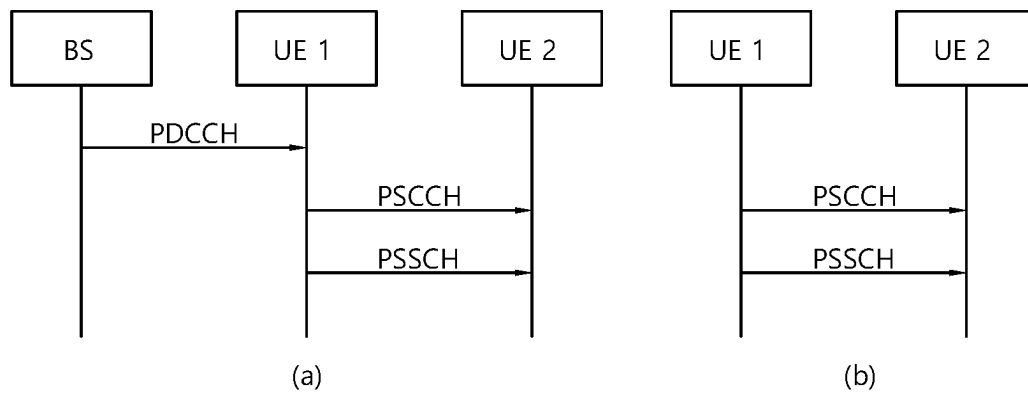


FIG. 9

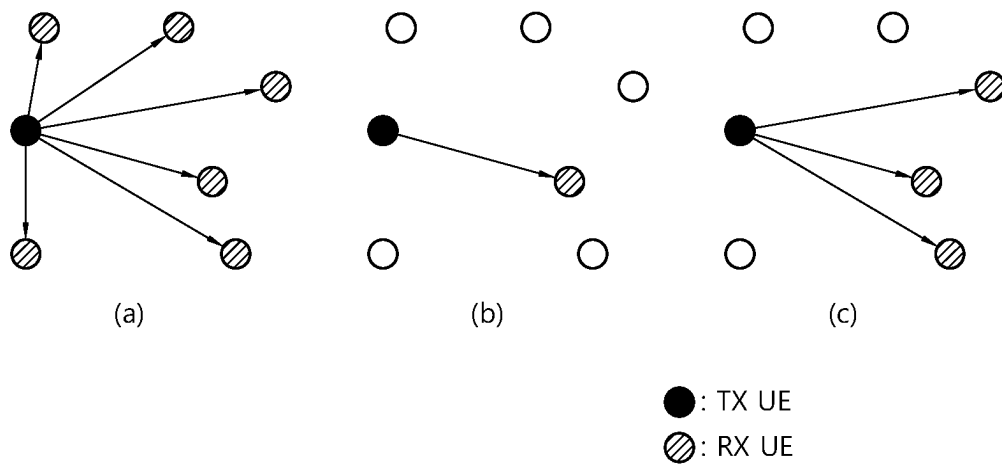


FIG. 10

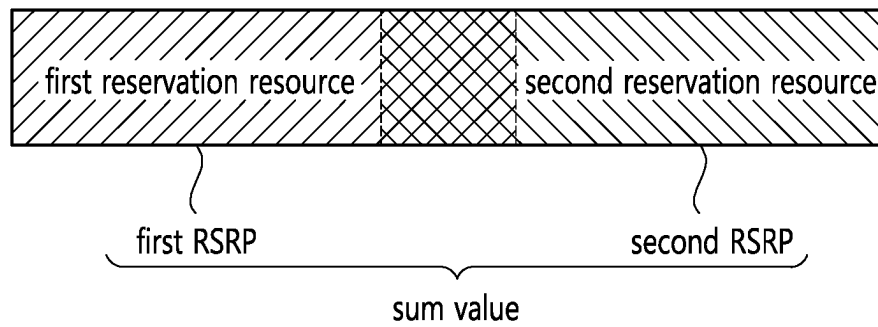


FIG. 11

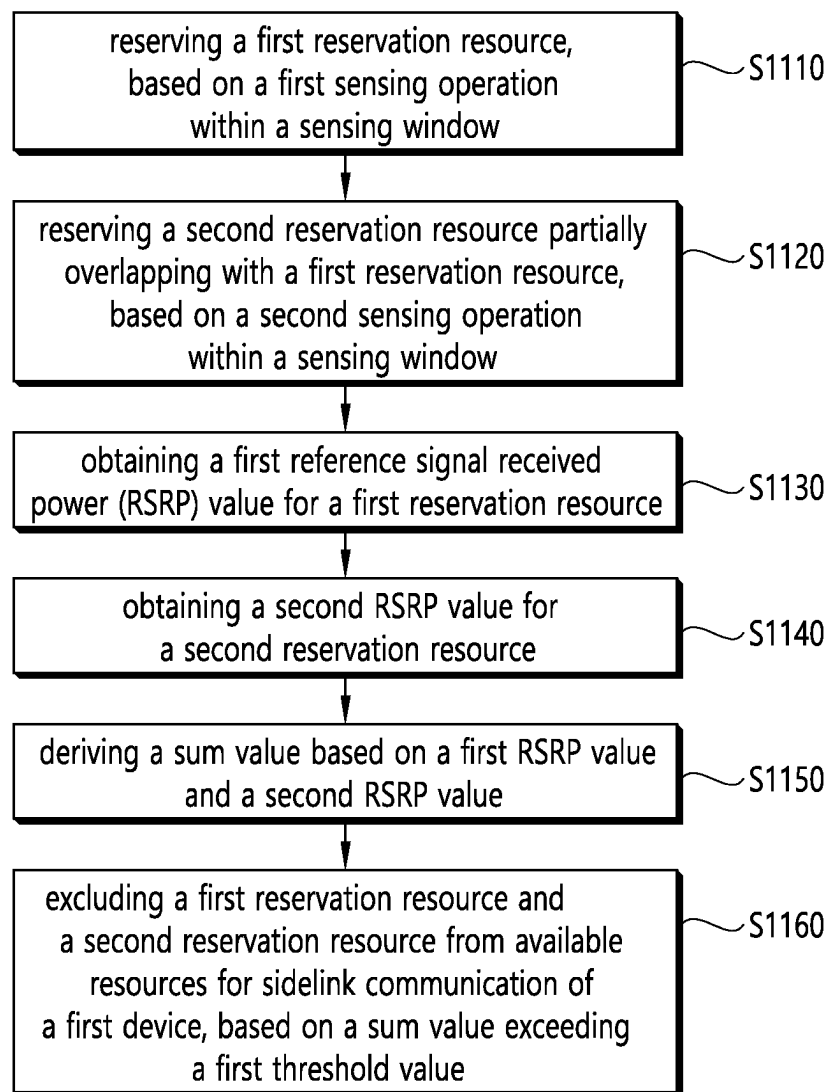


FIG. 12

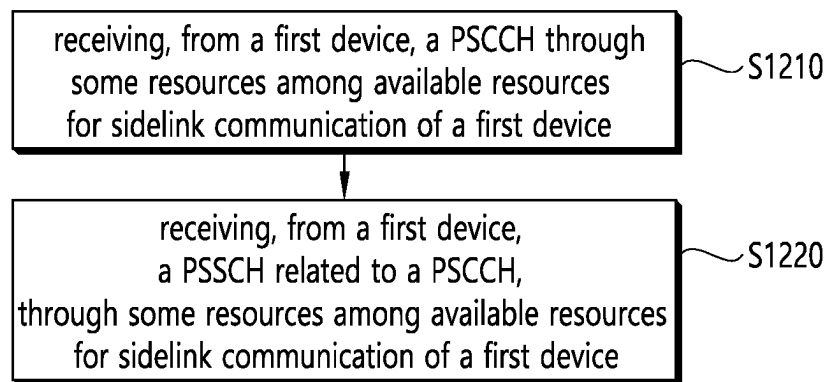


FIG. 13

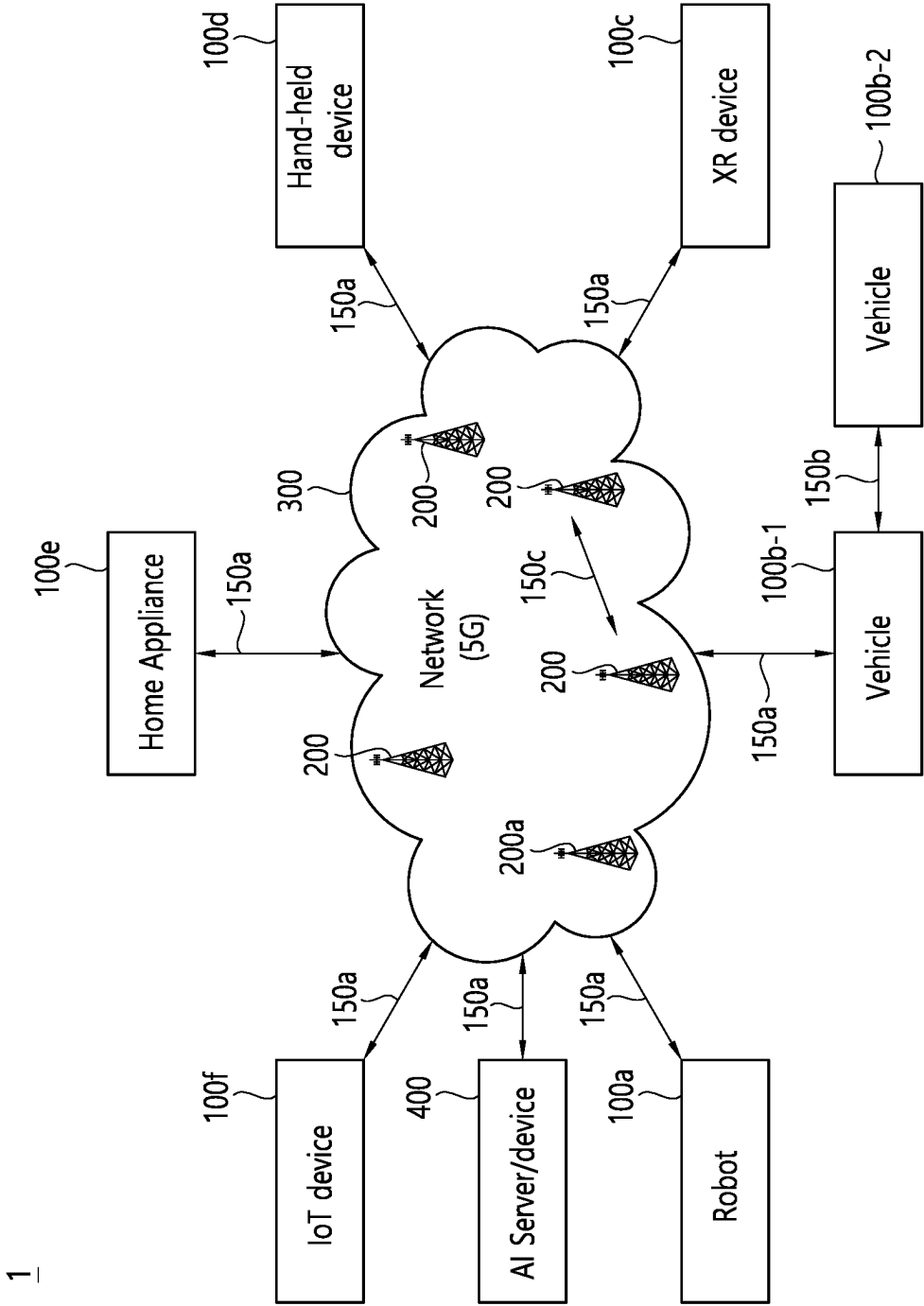


FIG. 14

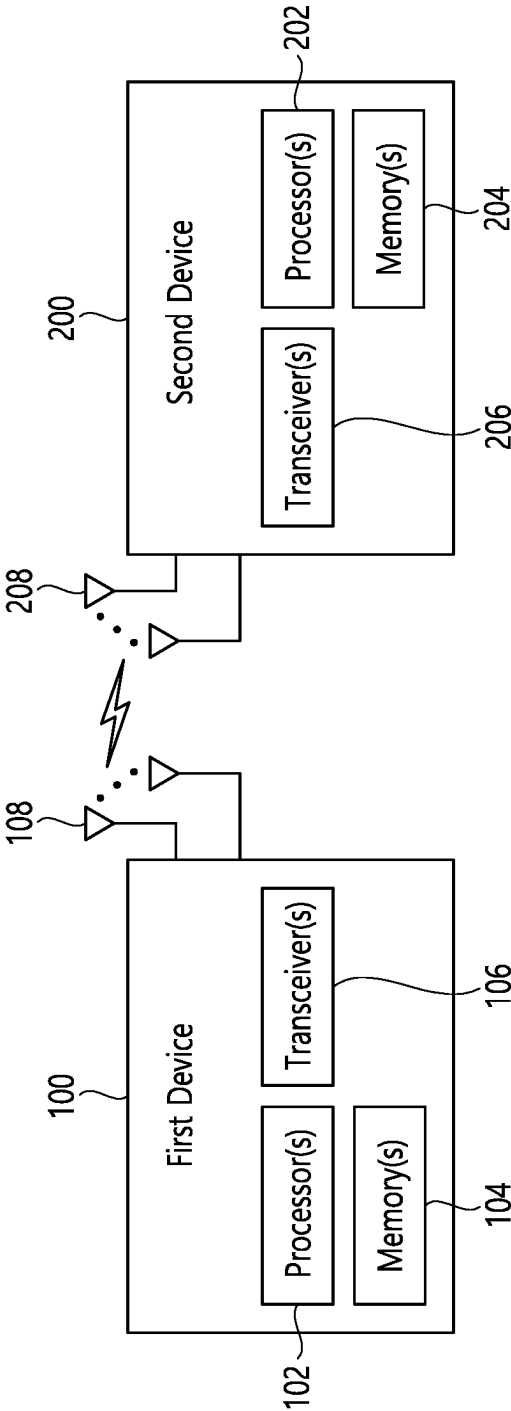


FIG. 15

1000(102/106, 202/206)

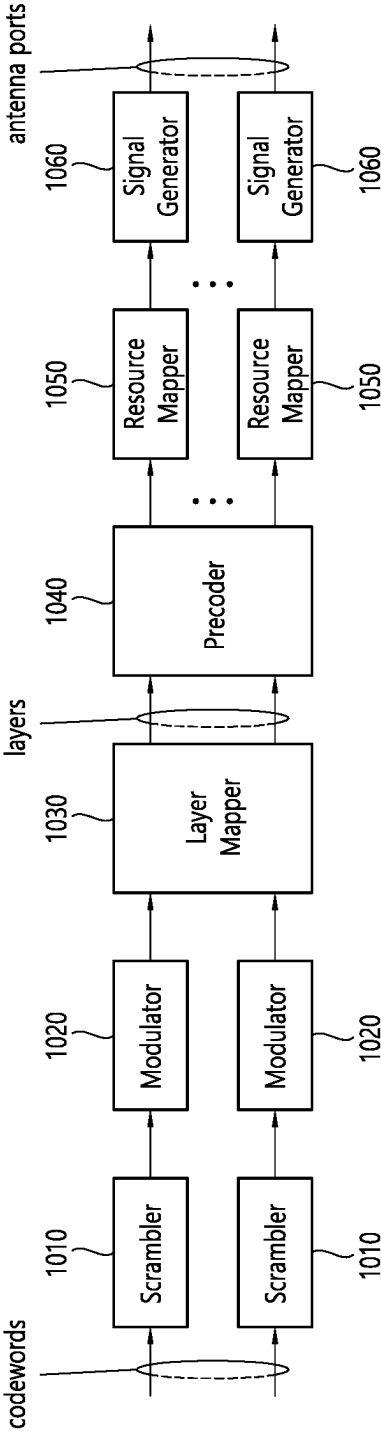


FIG. 16

Device (100,200)

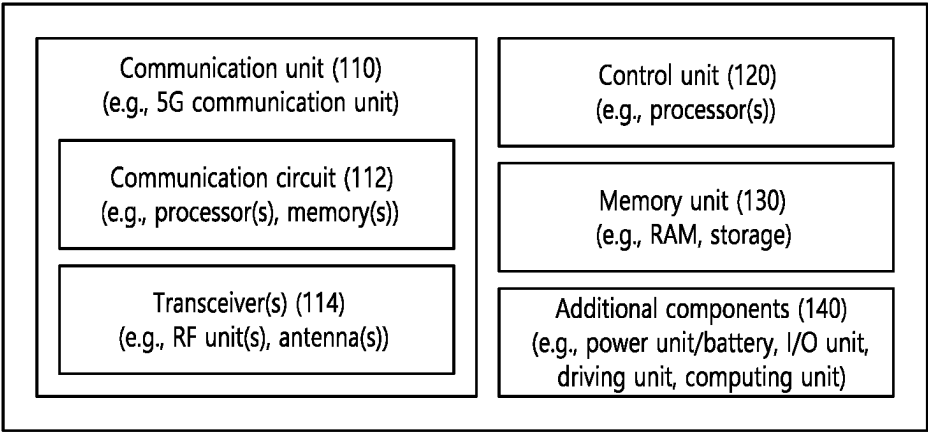


FIG. 17

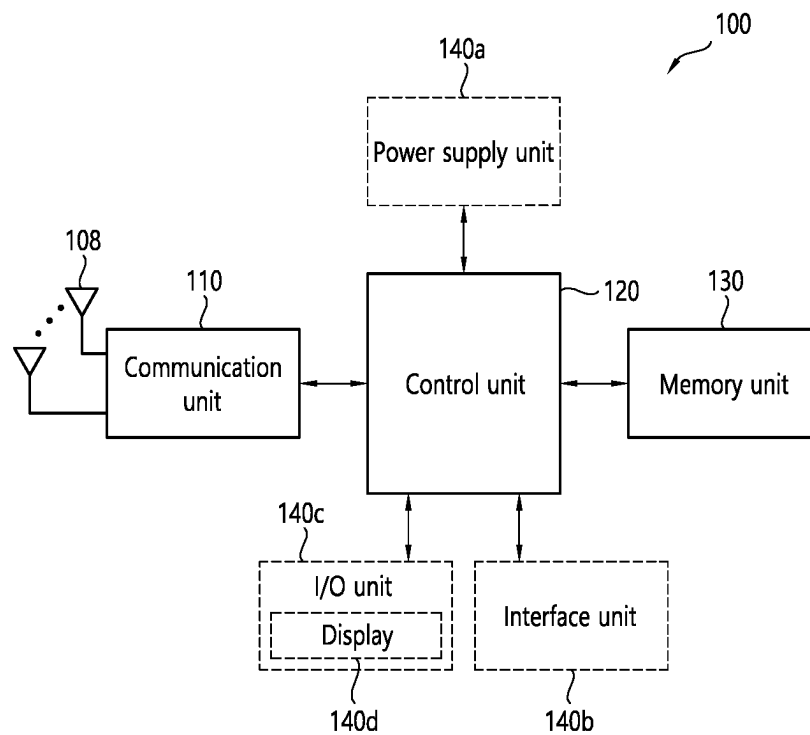
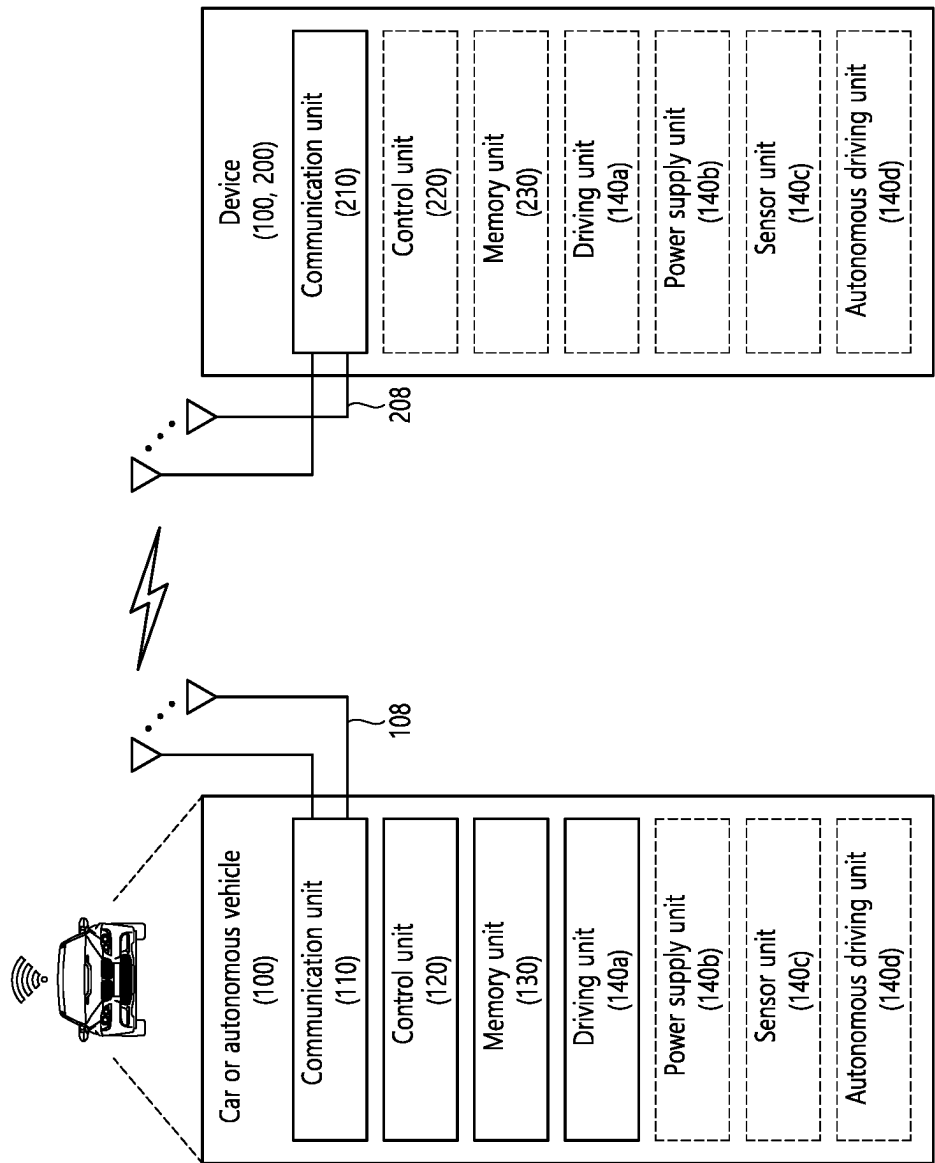


FIG. 18



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METHOD AND DEVICE FOR DETERMINING SIDELINK TRANSMISSION RESOURCE ON BASIS OF RSRP VALUE IN NR V2X

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2021/008708, filed on Jul. 8, 2021, which claims the benefit of earlier filing date and right of priority to Korean Application Nos. 10-2020-0095507, filed on Jul. 30, 2020, and 10-2020-0098810, filed on Aug. 6, 2020, the contents of which are all incorporated by reference herein in their entirety.

TECHNICAL FIELD

This disclosure relates to a wireless communication system.

BACKGROUND

Sidelink (SL) communication is a communication scheme in which a direct link is established between User Equipments (UEs) and the UEs exchange voice and data directly with each other without intervention of an evolved Node B (eNB). SL communication is under consideration as a solution to the overhead of an eNB caused by rapidly increasing data traffic. Vehicle-to-everything (V2X) refers to a communication technology through which a vehicle exchanges information with another vehicle, a pedestrian, an object having an infrastructure (or infra) established therein, and so on. The V2X may be divided into 4 types, such as vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P). The V2X communication may be provided via a PC5 interface and/or Uu interface.

Meanwhile, as a wider range of communication devices require larger communication capacities, the need for mobile broadband communication that is more enhanced than the existing Radio Access Technology (RAT) is rising. Accordingly, discussions are made on services and user equipment (UE) that are sensitive to reliability and latency. And, a next generation radio access technology that is based on the enhanced mobile broadband communication, massive Machine Type Communication (MTC), Ultra-Reliable and Low Latency Communication (URLLC), and so on, may be referred to as a new radio access technology (RAT) or new radio (NR). Herein, the NR may also support vehicle-to-everything (V2X) communication.

FIG. 1 is a drawing for describing V2X communication based on NR, compared to V2X communication based on RAT used before NR. The embodiment of FIG. 1 may be combined with various embodiments of the present disclosure.

Regarding V2X communication, a scheme of providing a safety service, based on a V2X message such as Basic Safety Message (BSM), Cooperative Awareness Message (CAM), and Decentralized Environmental Notification Message (DENM) is focused in the discussion on the RAT used before the NR. The V2X message may include position information, dynamic information, attribute information, or the like. For example, a UE may transmit a periodic message type CAM and/or an event triggered message type DENM to another UE.

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Thereafter, regarding V2X communication, various V2X scenarios are proposed in NR. For example, the various V2X scenarios may include vehicle platooning, advanced driving, extended sensors, remote driving, or the like.

SUMMARY

The technical object of the present disclosure is to propose a method and a device (or UE) for performing it, for performing sidelink (SL) communication between devices (or UEs).

Another technical object of the present disclosure is to propose a method and a device (or UE) for performing it, for determining a sidelink transmission resource based on an RSRP value in NR V2X.

According to an embodiment of the present disclosure, a method for performing, by a first device, sidelink communication may be proposed. The method may comprise: reserving a first reservation resource, based on a first sensing operation within a sensing window; reserving a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window; obtaining a first reference signal received power (RSRP) value for the first reservation resource; obtaining a second RSRP value for the second reservation resource; deriving a sum value based on the first RSRP value and the second RSRP value; and excluding the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

According to an embodiment of the present disclosure, a first device for performing sidelink communication may be proposed. The first device may comprise: at least one memory storing instructions; at least one transceiver; and at least one processor connected to the at least one memory and the at least one transceiver. For example, at least one processor may execute the instructions to: reserve a first reservation resource, based on a first sensing operation within a sensing window; reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window; obtain a first reference signal received power (RSRP) value for the first reservation resource; obtain a second RSRP value for the second reservation resource; derive a sum value based on the first RSRP value and the second RSRP value; and exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

A UE can efficiently perform SL communication.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a drawing for describing V2X communication based on NR, compared to V2X communication based on RAT used before NR.

FIG. 2 shows a structure of an NR system, in accordance with an embodiment of the present disclosure.

FIG. 3 shows a radio protocol architecture, in accordance with an embodiment of the present disclosure.

FIG. 4 shows a structure of a radio frame of an NR, in accordance with an embodiment of the present disclosure.

FIG. 5 shows a structure of a slot of an NR frame, in accordance with an embodiment of the present disclosure.

FIG. 6 shows an example of a BWP, in accordance with an embodiment of the present disclosure.

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FIG. 7 shows a UE performing V2X or SL communication, in accordance with an embodiment of the present disclosure.

FIG. 8 shows a procedure of performing V2X or SL communication by a UE based on a transmission mode, in accordance with an embodiment of the present disclosure.

FIG. 9 shows three cast types, in accordance with an embodiment of the present disclosure.

FIG. 10 is a figure for describing a method for determining available resources for sidelink communication based on a first RSRP value of a first reservation resource and a second RSRP value of a second reservation resource, according to an embodiment of the present disclosure.

FIG. 11 is a flowchart showing a method in which a first device performs sidelink communication according to an embodiment of the present disclosure.

FIG. 12 is a flowchart showing a method for a second device to perform sidelink communication according to an embodiment of the present disclosure.

FIG. 13 shows a communication system 1, in accordance with an embodiment of the present disclosure.

FIG. 14 shows wireless devices, in accordance with an embodiment of the present disclosure.

FIG. 15 shows a signal process circuit for a transmission signal, in accordance with an embodiment of the present disclosure.

FIG. 16 shows another example of a wireless device, in accordance with an embodiment of the present disclosure.

FIG. 17 shows a hand-held device, in accordance with an embodiment of the present disclosure.

FIG. 18 shows a vehicle or an autonomous vehicle, in accordance with an embodiment of the present disclosure.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

In the present specification, “A or B” may mean “only A”, “only B” or “both A and B”. In other words, in the present specification, “A or B” may be interpreted as “A and/or B”. For example, in the present specification, “A, B, or C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, C”.

A slash (/) or comma used in the present specification may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B, or C”.

In the present specification, “at least one of A and B” may mean “only A”, “only B”, or “both A and B”. In addition, in the present specification, the expression “at least one of A or B” or “at least one of A and/or B” may be interpreted as “at least one of A and B”.

In addition, in the present specification, “at least one of A, B, and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, and C”. In addition, “at least one of A, B, or C” or “at least one of A, B, and/or C” may mean “at least one of A, B, and C”.

In addition, a parenthesis used in the present specification may mean “for example”. Specifically, when indicated as “control information (PDCCH)”, it may mean that “PDCCH” is proposed as an example of the “control information”. In other words, the “control information” of the present specification is not limited to “PDCCH”, and “PDCCH” may be proposed as an example of the “control information”. In addition, when indicated as “control information (i.e., PDCCH)”, it may also mean that “PDCCH” is proposed as an example of the “control information”.

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A technical feature described individually in one figure in the present specification may be individually implemented, or may be simultaneously implemented.

The technology described below may be used in various wireless communication systems such as code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), orthogonal frequency division multiple access (OFDMA), single carrier frequency division multiple access (SC-FDMA), and so on. The CDMA may be implemented with a radio technology, such as universal terrestrial radio access (UTRA) or CDMA-2000. The TDMA may be implemented with a radio technology, such as global system for mobile communications (GSM)/general packet radio service (GPRS)/enhanced data rate for GSM evolution (EDGE). The OFDMA may be implemented with a radio technology, such as institute of electrical and electronics engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, evolved UTRA (E-UTRA), and so on. IEEE 802.16m is an evolved version of IEEE 802.16e and provides backward compatibility with a system based on the IEEE 802.16e. The UTRA is part of a universal mobile telecommunication system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is part of an evolved UMTS (E-UMTS) using the E-UTRA. The 3GPP LTE uses the OFDMA in a downlink and uses the SC-FDMA in an uplink. LTE-advanced (LTE-A) is an evolution of the LTE.

5G NR is a successive technology of LTE-A corresponding to a new Clean-slate type mobile communication system having the characteristics of high performance, low latency, high availability, and so on. 5G NR may use resources of all spectrum available for usage including low frequency bands of less than 1 GHz, middle frequency bands ranging from 1 GHz to 10 GHz, high frequency (millimeter waves) of 24 GHz or more, and so on.

For clarity in the description, the following description will mostly focus on LTE-A or 5G NR. However, technical features according to an embodiment of the present disclosure will not be limited only to this.

FIG. 2 shows a structure of an NR system, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 2 may be combined with various embodiments of the present disclosure.

Referring to FIG. 2, a next generation—radio access network (NG-RAN) may include a BS 20 providing a UE 10 with a user plane and control plane protocol termination. For example, the BS 20 may include a next generation-Node B (gNB) and/or an evolved-NodeB (eNB). For example, the UE 10 may be fixed or mobile and may be referred to as other terms, such as a mobile station (MS), a user terminal (UT), a subscriber station (SS), a mobile terminal (MT), wireless device, and so on. For example, the BS may be referred to as a fixed station which communicates with the UE 10 and may be referred to as other terms, such as a base transceiver system (BTS), an access point (AP), and so on.

The embodiment of FIG. 2 exemplifies a case where only the gNB is included. The BSs 20 may be connected to one another via Xn interface. The BS 20 may be connected to one another via 5th generation (5G) core network (5GC) and NG interface. More specifically, the BSs 20 may be connected to an access and mobility management function (AMF) 30 via NG-C interface, and may be connected to a user plane function (UPF) 30 via NG-U interface.

Layers of a radio interface protocol between the UE and the network can be classified into a first layer (layer 1, L1), a second layer (layer 2, L2), and a third layer (layer 3, L3)

based on the lower three layers of the open system interconnection (OSI) model that is well-known in the communication system. Among them, a physical (PHY) layer belonging to the first layer provides an information transfer service by using a physical channel, and a radio resource control (RRC) layer belonging to the third layer serves to control a radio resource between the UE and the network. For this, the RRC layer exchanges an RRC message between the UE and the BS.

FIG. 3 shows a radio protocol architecture, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 3 may be combined with various embodiments of the present disclosure. Specifically, (a) of FIG. 3 shows a radio protocol stack of a user plane for Uu communication, and (b) of FIG. 3 shows a radio protocol stack of a control plane for Uu communication. (c) of FIG. 3 shows a radio protocol stack of a user plane for SL communication, and (d) of FIG. 3 shows a radio protocol stack of a control plane for SL communication.

Referring to FIG. 3, a physical layer provides an upper layer with an information transfer service through a physical channel. The physical layer is connected to a medium access control (MAC) layer which is an upper layer of the physical layer through a transport channel. Data is transferred between the MAC layer and the physical layer through the transport channel. The transport channel is classified according to how and with what characteristics data is transmitted through a radio interface.

Between different physical layers, i.e., a physical layer of a transmitter and a physical layer of a receiver, data are transferred through the physical channel. The physical channel is modulated using an orthogonal frequency division multiplexing (OFDM) scheme, and utilizes time and frequency as a radio resource.

The MAC layer provides services to a radio link control (RLC) layer, which is a higher layer of the MAC layer, via a logical channel. The MAC layer provides a function of mapping multiple logical channels to multiple transport channels. The MAC layer also provides a function of logical channel multiplexing by mapping multiple logical channels to a single transport channel. The MAC layer provides data transfer services over logical channels.

The RLC layer performs concatenation, segmentation, and reassembly of Radio Link Control Service Data Unit (RLC SDU). In order to ensure diverse quality of service (QoS) required by a radio bearer (RB), the RLC layer provides three types of operation modes, i.e., a transparent mode (TM), an unacknowledged mode (UM), and an acknowledged mode (AM). An AM RLC provides error correction through an automatic repeat request (ARQ).

A radio resource control (RRC) layer is defined only in the control plane. The RRC layer serves to control the logical channel, the transport channel, and the physical channel in association with configuration, reconfiguration and release of RBs. The RB is a logical path provided by the first layer (i.e., the physical layer or the PHY layer) and the second layer (i.e., a MAC layer, an RLC layer, a packet data convergence protocol (PDCP) layer, and a service data adaptation protocol (SDAP) layer) for data delivery between the UE and the network.

Functions of a packet data convergence protocol (PDCP) layer in the user plane include user data delivery, header compression, and ciphering. Functions of a PDCP layer in the control plane include control-plane data delivery and ciphering/integrity protection.

A service data adaptation protocol (SDAP) layer is defined only in a user plane. The SDAP layer performs

mapping between a Quality of Service (QoS) flow and a data radio bearer (DRB) and QoS flow ID (QFI) marking in both DL and UL packets.

The configuration of the RB implies a process for specifying a radio protocol layer and channel properties to provide a particular service and for determining respective detailed parameters and operations. The RB can be classified into two types, i.e., a signaling RB (SRB) and a data RB (DRB). The SRB is used as a path for transmitting an RRC message in the control plane. The DRB is used as a path for transmitting user data in the user plane.

When an RRC connection is established between an RRC layer of the UE and an RRC layer of the E-UTRAN, the UE is in an RRC_CONNECTED state, and, otherwise, the UE may be in an RRC_IDLE state. In case of the NR, an RRC_INACTIVE state is additionally defined, and a UE being in the RRC_INACTIVE state may maintain its connection with a core network whereas its connection with the BS is released.

Data is transmitted from the network to the UE through a downlink transport channel. Examples of the downlink transport channel include a broadcast channel (BCH) for transmitting system information and a downlink-shared channel (SCH) for transmitting user traffic or control messages. Traffic of downlink multicast or broadcast services or the control messages can be transmitted on the downlink-SCH or an additional downlink multicast channel (MCH). Data is transmitted from the UE to the network through an uplink transport channel. Examples of the uplink transport channel include a random access channel (RACH) for transmitting an initial control message and an uplink SCH for transmitting user traffic or control messages.

Examples of logical channels belonging to a higher channel of the transport channel and mapped onto the transport channels include a broadcast channel (BCCH), a paging control channel (PCCH), a common control channel (CCCH), a multicast control channel (MCCH), a multicast traffic channel (MTCH), etc.

FIG. 4 shows a structure of a radio frame of an NR, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 4 may be combined with various embodiments of the present disclosure.

Referring to FIG. 4, in the NR, a radio frame may be used for performing uplink and downlink transmission. A radio frame has a length of 10 ms and may be defined to be configured of two half-frames (HFs). A half-frame may include five 1 ms subframes (SFs). A subframe (SF) may be divided into one or more slots, and the number of slots within a subframe may be determined in accordance with subcarrier spacing (SCS). Each slot may include 12 or 14 OFDM(A) symbols according to a cyclic prefix (CP).

In case of using a normal CP, each slot may include 14 symbols. In case of using an extended CP, each slot may include 12 symbols. Herein, a symbol may include an OFDM symbol (or CP-OFDM symbol) and a Single Carrier-FDMA (SC-FDMA) symbol (or Discrete Fourier Transform-spread-OFDM (DFT-s-OFDM) symbol).

Table 1 shown below represents an example of a number of symbols per slot ($N_{\text{slot}}^{\text{symb}}$), a number slots per frame ($N_{\text{frame},u}^{\text{slot}}$), and a number of slots per subframe ($N_{\text{subframe},u}^{\text{slot}}$) in accordance with an SCS configuration (u), in a case where a normal CP is used.

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TABLE 1

SCS ($15 \cdot 2^u$)	$N_{\text{slot}}^{\text{slot}}_{\text{symb}}$	$N_{\text{frame}, u}^{\text{frame}, u}_{\text{slot}}$	$N_{\text{subframe}, u}^{\text{subframe}, u}_{\text{slot}}$
15 KHz ($u = 0$)	14	10	1
30 KHz ($u = 1$)	14	20	2
60 KHz ($u = 2$)	14	40	4
120 KHz ($u = 3$)	14	80	8
240 KHz ($u = 4$)	14	160	16

Table 2 shows an example of a number of symbols per slot, a number of slots per frame, and a number of slots per subframe in accordance with the SCS, in a case where an extended CP is used.

TABLE 2

SCS ($15 \cdot 2^u$)	$N_{\text{slot}}^{\text{slot}}_{\text{symb}}$	$N_{\text{frame}, u}^{\text{frame}, u}_{\text{slot}}$	$N_{\text{subframe}, u}^{\text{subframe}, u}_{\text{slot}}$
60 KHz ($u = 2$)	12	40	4

In an NR system, OFDM(A) numerologies (e.g., SCS, CP length, and so on) between multiple cells being integrated to one UE may be differently configured. Accordingly, a (absolute time) duration (or section) of a time resource (e.g., subframe, slot or TTI) (collectively referred to as a time unit (TU) for simplicity) being configured of the same number of symbols may be differently configured in the integrated cells.

In the NR, multiple numerologies or SCSs for supporting diverse 5G services may be supported. For example, in case an SCS is 15 kHz, a wide area of the conventional cellular bands may be supported, and, in case an SCS is 30 kHz/60 kHz a dense-urban, lower latency, wider carrier bandwidth may be supported. In case the SCS is 60 kHz or higher, a bandwidth that is greater than 24.25 GHz may be used in order to overcome phase noise.

An NR frequency band may be defined as two different types of frequency ranges. The two different types of frequency ranges may be FR1 and FR2. The values of the frequency ranges may be changed (or varied), and, for example, the two different types of frequency ranges may be as shown below in Table 3. Among the frequency ranges that are used in an NR system, FR1 may mean a “sub 6 GHz range”, and FR2 may mean an “above 6 GHz range” and may also be referred to as a millimeter wave (mmW).

TABLE 3

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	450 MHz-6000 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

As described above, the values of the frequency ranges in the NR system may be changed (or varied). For example, as shown below in Table 4, FR1 may include a band within a range of 410 MHz to 7125 MHz. More specifically, FR1 may include a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, and so on) and higher. For example, a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, and so on) and higher being included in FR1 may include an unlicensed band. The unlicensed band may be used for diverse purposes, e.g., the unlicensed band for vehicle-specific communication (e.g., automated driving).

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TABLE 4

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

FIG. 5 shows a structure of a slot of an NR frame, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 5 may be combined with various embodiments of the present disclosure.

Referring to FIG. 5, a slot includes a plurality of symbols in a time domain. For example, in case of a normal CP, one slot may include 14 symbols. However, in case of an extended CP, one slot may include 12 symbols. Alternatively, in case of a normal CP, one slot may include 7 symbols. However, in case of an extended CP, one slot may include 6 symbols.

A carrier includes a plurality of subcarriers in a frequency domain. A Resource Block (RB) may be defined as a plurality of consecutive subcarriers (e.g., 12 subcarriers) in the frequency domain. A Bandwidth Part (BWP) may be defined as a plurality of consecutive (Physical) Resource Blocks ((P)RBs) in the frequency domain, and the BWP may correspond to one numerology (e.g., SCS, CP length, and so on). A carrier may include a maximum of N number BWPs (e.g., 5 BWPs). Data communication may be performed via an activated BWP. Each element may be referred to as a Resource Element (RE) within a resource grid and one complex symbol may be mapped to each element.

Hereinafter, a bandwidth part (BWP) and a carrier will be described.

The BWP may be a set of consecutive physical resource blocks (PRBs) in a given numerology. The PRB may be selected from consecutive sub-sets of common resource blocks (CRBs) for the given numerology on a given carrier.

For example, the BWP may be at least any one of an active BWP, an initial BWP, and/or a default BWP. For example, the UE may not monitor downlink radio link quality in a DL BWP other than an active DL BWP on a primary cell (PCell). For example, the UE may not receive PDCCH, physical downlink shared channel (PDSCH), or channel state information-reference signal (CSI-RS) (excluding RRM) outside the active DL BWP. For example, the UE may not trigger a channel state information (CSI) report for the inactive DL BWP. For example, the UE may not transmit physical uplink control channel (PUCCH) or physical uplink shared channel (PUSCH) outside an active UL BWP. For example, in a downlink case, the initial BWP may be given as a consecutive RB set for a remaining minimum system information (RMSI) control resource set (CORESET) (configured by physical broadcast channel (PBCH)). For example, in an uplink case, the initial BWP may be given by system information block (SIB) for a random access procedure. For example, the default BWP may be configured by a higher layer. For example, an initial value of the default BWP may be an initial DL BWP. For energy saving, if the UE fails to detect downlink control information (DCI) during a specific period, the UE may switch the active BWP of the UE to the default BWP.

Meanwhile, the BWP may be defined for SL. The same SL BWP may be used in transmission and reception. For example, a transmitting UE may transmit an SL channel or an SL signal on a specific BWP, and a receiving UE may receive the SL channel or the SL signal on the specific BWP. In a licensed carrier, the SL BWP may be defined separately

from a Uu BWP, and the SL BWP may have configuration signaling separate from the Uu BWP. For example, the UE may receive a configuration for the SL BWP from the BS/network. For example, the UE may receive a configuration for the Uu BWP from the BS/network. The SL BWP may be (pre-)configured in a carrier with respect to an out-of-coverage NR V2X UE and an RRC_IDLE UE. For the UE in the RRC_CONNECTED mode, at least one SL BWP may be activated in the carrier.

FIG. 6 shows an example of a BWP, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 6 may be combined with various embodiments of the present disclosure. It is assumed in the embodiment of FIG. 6 that the number of BWPs is 3.

Referring to FIG. 6, a common resource block (CRB) may be a carrier resource block numbered from one end of a carrier band to the other end thereof. In addition, the PRB may be a resource block numbered within each BWP. A point A may indicate a common reference point for a resource block grid.

The BWP may be configured by a point A, an offset N_{BWP}^{start} from the point A, and a bandwidth N_{BWP}^{size} . For example, the point A may be an external reference point of a PRB of a carrier in which a subcarrier 0 of all numerologies (e.g., all numerologies supported by a network on that carrier) is aligned. For example, the offset may be a PRB interval between a lowest subcarrier and the point A in a given numerology. For example, the bandwidth may be the number of PRBs in the given numerology.

Hereinafter, V2X or SL communication will be described.

A sidelink synchronization signal (SLSS) may include a primary sidelink synchronization signal (PSSS) and a secondary sidelink synchronization signal (SSSS), as an SL-specific sequence. The PSSS may be referred to as a sidelink primary synchronization signal (S-PSS), and the SSSS may be referred to as a sidelink secondary synchronization signal (S-SSS). For example, length-127 M-sequences may be used for the S-PSS, and length-127 gold sequences may be used for the S-SSS. For example, a UE may use the S-PSS for initial signal detection and for synchronization acquisition. For example, the UE may use the S-PSS and the S-SSS for acquisition of detailed synchronization and for detection of a synchronization signal ID.

A physical sidelink broadcast channel (PSBCH) may be a (broadcast) channel for transmitting default (system) information which must be first known by the UE before SL signal transmission/reception. For example, the default information may be information related to SLSS, a duplex mode (DM), a time division duplex (TDD) uplink/downlink (UL/DL) configuration, information related to a resource pool, a type of an application related to the SLSS, a subframe offset, broadcast information, or the like. For example, for evaluation of PSBCH performance, in NR V2X, a payload size of the PSBCH may be 56 bits including 24-bit cyclic redundancy check (CRC).

The S-PSS, the S-SSS, and the PSBCH may be included in a block format (e.g., SL synchronization signal (SS)/PSBCH block, hereinafter, sidelink-synchronization signal block (S-SSB)) supporting periodical transmission. The S-SSB may have the same numerology (i.e., SCS and CP length) as a physical sidelink control channel (PSCCH)/physical sidelink shared channel (PSSCH) in a carrier, and a transmission bandwidth may exist within a (pre-)configured sidelink (SL) BWP. For example, the S-SSB may have a bandwidth of 11 resource blocks (RBs). For example, the PSBCH may exist across 11 RBs. In addition, a frequency position of the S-SSB may be (pre-)configured. Accordingly,

the UE does not have to perform hypothesis detection at frequency to discover the S-SSB in the carrier.

FIG. 7 shows a UE performing V2X or SL communication, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 7 may be combined with various embodiments of the present disclosure.

Referring to FIG. 7, in V2X or SL communication, the term 'UE' may generally imply a UE of a user. However, if a network equipment such as a BS transmits/receives a signal according to a communication scheme between UEs, the BS may also be regarded as a sort of the UE. For example, a UE 1 may be a first apparatus 100, and a UE 2 may be a second apparatus 200.

For example, the UE 1 may select a resource unit corresponding to a specific resource in a resource pool which implies a set of series of resources. In addition, the UE 1 may transmit an SL signal by using the resource unit. For example, a resource pool in which the UE 1 is capable of transmitting a signal may be configured to the UE 2 which is a receiving UE, and the signal of the UE 1 may be detected in the resource pool.

Herein, if the UE 1 is within a connectivity range of the BS, the BS may inform the UE 1 of the resource pool. Otherwise, if the UE 1 is out of the connectivity range of the BS, another UE may inform the UE 1 of the resource pool, or the UE 1 may use a pre-configured resource pool.

In general, the resource pool may be configured in unit of a plurality of resources, and each UE may select a unit of one or a plurality of resources to use it in SL signal transmission thereof.

Hereinafter, resource allocation in SL will be described.

FIG. 8 shows a procedure of performing V2X or SL communication by a UE based on a transmission mode, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 8 may be combined with various embodiments of the present disclosure. In various embodiments of the present disclosure, the transmission mode may be called a mode or a resource allocation mode. Hereinafter, for convenience of explanation, in LTE, the transmission mode may be called an LTE transmission mode. In NR, the transmission mode may be called an NR resource allocation mode.

For example, (a) of FIG. 8 shows a UE operation related to an LTE transmission mode 1 or an LTE transmission mode 3. Alternatively, for example, (a) of FIG. 8 shows a UE operation related to an NR resource allocation mode 1. For example, the LTE transmission mode 1 may be applied to general SL communication, and the LTE transmission mode 3 may be applied to V2X communication.

For example, (b) of FIG. 8 shows a UE operation related to an LTE transmission mode 2 or an LTE transmission mode 4. Alternatively, for example, (b) of FIG. 8 shows a UE operation related to an NR resource allocation mode 2.

Referring to (a) of FIG. 8, in the LTE transmission mode 1, the LTE transmission mode 3, or the NR resource allocation mode 1, a BS may schedule an SL resource to be used by the UE for SL transmission. For example, the BS may perform resource scheduling to a UE 1 through a PDCCH (e.g., downlink control information (DCI)) or RRC signaling (e.g., Configured Grant Type 1 or Configured Grant Type 2), and the UE 1 may perform V2X or SL communication with respect to a UE 2 according to the resource scheduling. For example, the UE 1 may transmit a sidelink control information (SCI) to the UE 2 through a physical sidelink control channel (PSCCH), and thereafter transmit data based on the SCI to the UE 2 through a physical sidelink shared channel (PSSCH).

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Referring to (b) of FIG. 8, in the LTE transmission mode 2, the LTE transmission mode 4, or the NR resource allocation mode 2, the UE may determine an SL transmission resource within an SL resource configured by a BS/network or a pre-configured SL resource. For example, the configured SL resource or the pre-configured SL resource may be a resource pool. For example, the UE may autonomously select or schedule a resource for SL transmission. For example, the UE may perform SL communication by autonomously selecting a resource within a configured resource pool. For example, the UE may autonomously select a resource within a selective window by performing a sensing and resource (re)selection procedure. For example, the sensing may be performed in unit of subchannels. In addition, the UE 1 which has autonomously selected the resource within the resource pool may transmit the SCI to the UE 2 through a PSCCH, and thereafter may transmit data based on the SCI to the UE 2 through a PSSCH.

FIG. 9 shows three cast types, in accordance with an embodiment of the present disclosure. The embodiment of FIG. 9 may be combined with various embodiments of the present disclosure. Specifically, FIG. 9(a) shows broadcast-type SL communication, FIG. 9(b) shows unicast type-SL communication, and FIG. 9(c) shows groupcast-type SL communication. In case of the unicast-type SL communication, a UE may perform one-to-one communication with respect to another UE. In case of the groupcast-type SL transmission, the UE may perform SL communication with respect to one or more UEs in a group to which the UE belongs. In various embodiments of the present disclosure, SL groupcast communication may be replaced with SL multicast communication, SL one-to-many communication, or the like.

FIG. 10 is a figure for describing a method for determining available resources for sidelink communication based on a first RSRP value of a first reservation resource and a second RSRP value of a second reservation resource, according to an embodiment of the present disclosure.

Meanwhile, a UE may select SL resource within a resource pool based on sensing. In the description, a scheme where a UE selects an SL resource may be referred to mode 2 scheme, mode 2 operation or resource allocation mode 2. For example, in resource allocation mode 2, a UE may detect SCI transmitted by another UE, and the UE may obtain RSRP measurement value. And, a UE may select a resource to be used for SL transmission excluding a specific resource within a resource (re)selection window based on the above-described sensing result. In the case of the sensing operation, a UE may refer to resource allocation information received through first SCI.

Meanwhile, a UE may derive/determine a reservation resource based on information received through different SCI and/or different PSCCH and/or different PSSCH within a sensing window. Here, the reservation resource derived/determined by the UE based on different SCI and/or different PSCCH and/or different PSSCH may overlap in whole or in part within a resource (re)selection window. In the above case, the UE may predict/determine that a plurality of PSCCHs and/or a plurality of PSSCHs are overlapped and transmitted on the reservation resource. For example, in order for a UE to measure the interference level for each PSCCH and/or each PSSCH, the UE may measure RSRP based on a PSCCH and/or PSSCH indicating the reservation resource. Specifically, for example, a UE may measure RSRP based on a DMRS on a PSCCH through which information related to the reservation resource is transmitted and/or a DMRS on a PSSCH through which information

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related to the reservation resources is transmitted. In this case, when the RSRP value for the reservation resource exceeds a threshold value, a UE may exclude the reservation resource from available resources available for transmission. For example, the threshold value may be determined/configured according to a combination of a priority value indicated through the PSCCH and a priority value corresponding to data or packets to be transmitted by a UE.

For example, it is assumed that a first reservation resource and a second reservation resource overlap with each resource (eg, a combination of a subchannel and a slot). At this time, a UE may compare the RSRP measurement value for the first reservation resource with a first threshold value, and the UE may compare the RSRP measurement value for the second reservation resource with a second threshold value. Here, the first threshold value may be determined based on a priority value corresponding to the first reservation resource and a priority value of transmission data of a UE, and the second threshold value may be determined based on a priority value corresponding to the second reservation resource and a priority value for transmission data of a UE. For example, if the RSRP measurement value for a first reservation resource exceeds a first threshold value and/or the RSRP measurement value for a second reservation resource exceeds a second threshold value, a UE may exclude the first reservation resource and/or the second reservation resource from available resources for transmission.

On the other hand, even though the RSRP measurement value for a first reservation resource is smaller than a first threshold value and the RSRP measurement value for a second reservation resource is smaller than a second threshold value, a UE may need to exclude the first reservation resource and the second reservation resource from available resources for transmission. In other words, even if the interference level of each of a first reservation resource and a second reservation resource is low, the sum of the interference levels for the first reservation resource and the second reservation resource may still be high. In this case, it may be efficient for a UE to perform resource (re)selection based on the sum of interference levels for the first reservation resource and the second reservation resource.

Referring to FIG. 10, a UE may obtain the RSRP measurement value for a first reservation resource (first RSRP in FIG. 10) and the RSRP measurement value for a second reservation resource (second RSRP in FIG. 10). In addition, a UE may obtain/determine/derive a specific value ('sum value' in FIG. 10) based on the RSRP measurement value for the first reservation resource and the RSRP measurement value for the second reservation resource. For example, the specific value may be the sum of the RSRP measurement value for the first reservation resource and the RSRP measurement value for the second reservation resource. For example, the specific value may be an average of the RSRP measurement value for the first reservation resource and the RSRP measurement value for the second reservation resource. For example, the specific value may be a weighted average of the RSRP measurement value for the first reservation resource and the RSRP measurement value for the second reservation resource. For example, when the specific value exceeds a third threshold value, a UE may exclude the first reservation resource and/or the second reservation resource from available resources for transmission within the resource (re)selection window.

For example, if the RSRP measurement value for a first reservation resource exceeds a first threshold value, and the RSRP measurement value for a second reservation resource

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exceeds a second threshold value, and a specific value exceeds a third threshold value, a UE may exclude the first reservation resource and/or the second reservation resource from available resources for transmission within the resource (re)selection window.

For example, when the specific value exceeds a third threshold value without considering a first threshold value and a second threshold value, a UE may exclude the first reservation resource and/or the second reservation resource from available resources for transmission within the resource (re)selection window.

For example, the threshold value may be configured or pre-configured for a UE for each resource pool. For example, the threshold value may be configured or pre-configured for a UE for each combination of a priority value for a first reservation resource and/or a priority value for a second reservation resource and/or a priority value for data or packets to be transmitted. For example, the threshold value may be configured or pre-configure for a UE for each combination of a priority value for a first reservation resource and a minimum value for a priority value for a second reservation resource and a priority value for data or packets to be transmitted. For example, the threshold value may be configured or pre-configured for a UE for each combination of a priority value for a first reservation resource and a maximum value for a priority value for a second reservation resource and a priority value for data or packets to be transmitted. For example, the threshold value may be a third threshold value.

In the above-described embodiment, the situation in which two different reservation resources overlap has been described, but the technical idea of the present disclosure is not limited thereto. Even in a situation where three or more different reservation resources overlap, the technical idea of the present disclosure can be extended and applied.

Meanwhile, when a UE selects a resource to be used for transmission, the UE needs to consider the transmission power level as well as the interference level of a resource. For example, this is because a receiving UE can receive a PSCCH and/or a PSSCH through a first resource when the transmission power is high even when the interference level of the first resource is high. On the other hand, even if the interference level of a second resource is low, if the transmission power is low, this is because a receiving UE may not be able to receive a PSCCH and/or a PSSCH through the second resource.

For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be different for each range of power values per physical resource block (PRB) that a UE will use for transmission. For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be configured or pre-configured differently for a UE for each range of power values per PRB that the UE will use for transmission. For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be different for each combination of transmission and reception priority values and for each range of power values per PRB to be used for transmission by a UE. For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be configured or pre-configured differently for a UE for each combination of transmission and reception priority values and for each range of power values per PRB that the UE will use for transmission.

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For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be different for each power offset (nominal power (pre-)configured according to the path loss type) value range to be used by a UE for transmission. For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be configured or pre-configured differently for a UE for each power offset (nominal power (pre-)configured according to the path loss type) value range to be used by the UE for transmission.

For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be different for each combination of transmission and reception priority values and for each value range of the power offset (nominal power configured or (pre-)configured according to the path loss type) used by a UE for transmission. For example, a threshold value used for comparison with the RSRP measurement value corresponding to a reservation resource may be configured or pre-configured for a UE differently for each combination of transmission and reception priority values and for each value range of the power offset (nominal power configured or (pre-)configured according to the path loss type) used by the UE for transmission.

For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be different for each power control scheme which a UE uses for a transmission. For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be configured or pre-configured for a UE differently for each power control scheme which a UE uses for a transmission. For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be different for each combination of transmission and reception priority values and a power control scheme which a UE uses for a transmission. For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be configured or pre-configured for a UE differently for each combination of transmission and reception priority values and a power control scheme which a UE uses for a transmission. For example, a power control scheme may be at least one among a power control scheme when a DL path loss-based power control is enabled, a power control scheme when an SL path loss-based power control is enabled, a power control scheme when both DL path loss-based power control and SL path loss-based power control are enabled, and/or power control scheme when path loss is not considered.

For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be different for each UE type. For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be configured or pre-configured for a UE differently for each UE type. For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be different for each combination of transmission and reception priority values and UE type. For example, a threshold value used for comparison with a RSRP measurement value corresponding to a reservation resource may be configured or pre-configured for a UE differently for each combination of transmission and reception priority values and UE type. For example, UE type may be at least one among a UE operating in power saving mode,

a vehicle UE (V-UE), a pedestrian UE (P-UE), a road side unit (RSU), and/or a vulnerable road user (VRU).

Meanwhile, after resource (re)selection, the range of transmit power may be changed. For example, for simultaneous transmission of a UL channel and an SL channel, a UE may reduce transmit power for the SL channel. For example, the range of transmission power values used in resource (re)selection may assume that only an SL channel is transmitted. For example, when the actual transmission power value is changed to more than a specific threshold value, a UE may trigger resource (re)selection. For example, the specific threshold value may be configured or pre-configured for a UE for each resource pool. For example, the threshold value may be configured or pre-configure for a UE for each priority of transmission data. For example, the threshold value may be configured or pre-configure for a UE for each power saving mode.

For example, in resource allocation mode 2, a UE may determine whether to exclude a reservation resource from available resources within a resource (re)selection window using the ratio of a reference transmission power value and the RSRP measurement value for the reservation resource. For example, when the ratio value is less than a threshold value, a UE may exclude the reservation resource from available resources within a resource (re)selection window. For example, when the ratio value is less than or equal to a threshold value, a UE may exclude the reservation resource from available resources within a resource (re)selection window. For example, the threshold value may be configured or pre-configure for a UE for each priority value for combination of a reservation resource and a priority value corresponding to data to be transmitted.

For example, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources (and/or after a UE performs an operation to change (increase) an RSRP threshold value for the purpose of securing available resources above a pre-configure threshold value), when the ratio of available resources to total resources within a resource (re)selection window is greater than or equal to a (pre)configured threshold value, a UE may further apply various embodiments of the present disclosure. That is, various embodiments of the present disclosure may be applied when a UE selects a reservation resource or a candidate for a corresponding UE (reported/transmitted to a MAC layer by a PHY layer) within available resources. For example, even after a UE applies the additional resource exclusion method according to the embodiment, the UE may perform the operation according to various embodiments of the present disclosure only when the ratio to available resources is equal to or greater than a (pre)configured threshold value. For example, according to the ratio to the available resources, a UE may perform a change (increase) operation on an RSRP threshold value after applying the embodiment of the present disclosure. Thereafter, a UE may perform a process of selecting available resources again.

For example, after a UE applies the additional resource exclusion method according to the embodiment, even when the ratio to the available resources is equal to or less than (or less than) a (pre-)configured threshold value, the UE may perform operations according to various embodiments of the present disclosure. In the above case, for example, even when the ratio to the available resources is equal to or less than (or less than) a (pre-)configured threshold value, a UE may omit the process of increasing the corresponding threshold value and the process of reselecting available resources. That is, after a UE performs an operation of

excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, if the ratio of available resources is equal to or less than (or less than) a threshold value, the UE may increase the RSRP threshold value (eg, 3 [dB]), and the UE may reselect available resources based on the increased RSRP threshold value. On the other hand, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, after the ratio of available resources is equal to or greater than (or greater than) a threshold value and the UE applies the additional resource exclusion method according to the embodiment, even when the ratio of available resources is equal to or less than (or less than) a (pre-) configured threshold value, the UE may not increase an RSRP threshold value. That is, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, only when the ratio of available resources is equal to or less than (or less than) a threshold value, the UE may increase the RSRP threshold value (eg, 3 [dB]). On the other hand, after a UE applies the additional resource exclusion method according to the embodiment, even if the ratio of available resources is equal to or less than (or less than) a (pre)configured threshold value, the UE may not increase the RSRP threshold value.

Based on an embodiment of the present disclosure, a priority may be configured for each available resource or each available resource group, the UE may perform selection or exclusion of available resources according to the priority order until the ratio to the available resources becomes equal to or greater than the threshold value. For example, the threshold value for the available resource ratio may be the same value as the threshold value used after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources. For example, a threshold value for the available resource ratio may be a second threshold value different from a first threshold value used after a UE performs an operation of excluding available resources within an RSRP-based resource (re)selection window for previous reservation resources. In this case, the first threshold value and/or the second threshold value may be configured or pre-configure for a UE. That is, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, if the ratio of available resources is equal to or less than (or less than) a first threshold value, the UE may increase the first RSRP threshold value (eg, 3 [dB]), and the UE may reselect available resources based on the increased first RSRP threshold value. On the other hand, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, the ratio of available resources may be equal to or greater than (or greater than) a first threshold value. Here, if the ratio of available resources is equal to or greater than (or greater than) a first threshold value, a UE may apply an additional resource exclusion method according to an embodiment. After a UE applies the additional resource exclusion method, if the ratio of available resources is equal to or less than (or less than) a second threshold value (pre-)configured separately, the UE may increase the second RSRP threshold value (eg, 3 [dB]), the UE may reselect available resources based on the increased second RSRP threshold value. When the ratio of available resources is equal to or greater than (or greater than) a first threshold value after a UE performs an

operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, and the ratio of the available resources is equal to or greater than (or greater than) a second threshold value (pre-)configured separately after the UE applies the additional resource exclusion method according to the embodiment, the UE may not increase the RSRP threshold value. Here, for example, the first RSRP threshold value and the second RSRP threshold value may be different. For example, the first RSRP threshold value and the second RSRP threshold value may be the same.

For example, the priority for each available resource may be configured higher in the order of increasing RSRP sum values for reservation resources. For example, the priority for each available resource may be configured higher in order of smaller RSRP sum values for reservation resources. For example, The priority of each available resource may be configured to higher in order of increasing reference values corresponding to power corresponding to transmission data. For example, the priority of each available resource may be configured higher in order of decreasing reference values corresponding to power corresponding to transmission data. In the above application method, the process of comparing a threshold value with a new reference value in the embodiment of the present disclosure may be omitted, and the priority may be determined using the new reference value itself.

For example, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, after the ratio of available resources is equal to or greater than (or greater than) a first threshold value and the UE applies the additional resource exclusion method according to the embodiment, when the ratio of the available resources is equal to or less than (or less than) a second threshold value (pre-)configured separately, the UE may randomly select resources to be excluded from the additional resources according to the embodiment or select them according to the priority order. Accordingly, a UE may perform additional resource exclusion only until the ratio of available resources is equal to or less than (or less than) a first threshold value and/or equal to or greater than (or greater than) a second threshold value. Here, a first threshold value and a second threshold value may be different, and a first threshold value may be greater than a second threshold value.

For example, after a UE performs an operation of excluding available resources within a resource (re)selection window based on RSRP for previous reservation resources, after the ratio of available resources is equal to or greater than (or greater than) a first threshold value and the UE applies the additional resource exclusion method according to the embodiment, when the ratio of the available resources is equal to or less than (or less than) a second threshold value (pre-)configured separately, the UE may randomly select resources to be excluded from the additional resources according to the embodiment or select them according to the priority order. Accordingly, a UE may perform additional resource exclusion only until the ratio of available resources is equal to or greater than (or greater than) a first threshold value and/or equal to or greater than (or greater than) a second threshold value. Here, a first threshold value and a second threshold value may be the same.

For example, only when resource (re)selection is triggered, a UE may perform operations according to various embodiments of the present disclosure. Characteristically, for example, the case where the resource (re)selection is triggered may not include a case in which a UE reselects

resources by performing continuous sensing after performing resource (re)selection (e.g., RE-EVALUATION). That is, a case in which a UE performs resource (re)selection based on a pre-emption after performing resource (re)selection may be excluded from cases in which a resource (re)selection is triggered.

For example, when RE-EVALUATION-based resource (re)selection is triggered, a UE may perform operations according to various embodiments of the present disclosure. That is, when the reservation resource and/or its candidate group needs to be updated due to additional SCI detection, a UE may perform operations according to various embodiments of the present disclosure. For example, when a UE performs resource (re)selection based on a preemption, the UE may perform operations according to various embodiments of the present disclosure. That is, when a reservation resource and/or a candidate group for the reservation resource needs to be updated according to a pre-emption condition, a UE may perform an operation according to various embodiments of the present disclosure. For example, when a reception priority value is equal to or greater than a specific threshold value, the pre-emption condition may be met. For example, when a reception priority value is equal to or less than a specific threshold value, the pre-emption condition may be met. For example, when a transmission priority value is equal to or greater than a reception priority, the pre-emption condition may be met. For example, when a transmission priority value is equal to or less than a reception priority, the pre-emption condition may be met. For example, when a measurement RSRP value is greater than or equal to a threshold value, the pre-emption condition may be met. For example, when a measurement RSRP value is equal to or less than a threshold value, the pre-emption condition may be met.

For example, available resources and/or reservation resources and/or candidate groups for reservation resources configured according to various embodiments of the present disclosure may correspond to a specific transmission state. For example, the specific transmission state may correspond to initial transmission from the perspective of a transmitting UE for each TB. For example, the specific transmission state may correspond to initial transmission from the perspective of a receiving UE for each TB. In the above case, if a transmitting UE does not receive a PSFCH from a receiving UE (in unicast and/or groupcast option 2), or if a transmitting UE determines that a receiving UE has failed to decode SCI, the transmitting UE may configure the next retransmission as an initial transmission from the viewpoint of the receiving UE. For example, the specific transmission state may correspond to retransmission for each TB (eg, transmission other than the initial transmission for the same TB). For example, the specific transmission state may correspond to initial transmission and/or retransmission for a case where SL HARQ feedback is disabled (according to (pre)configuration and/or SCI indication) for each TB. For example, the specific transmission state may correspond to initial transmission and/or retransmission when SL HARQ feedback is activated for each TB (according to (pre)configuration and/or SCI indication). For example, the specific transmission state may correspond to transmission performed on a resource not included in previously reservation resources by a transmitting UE. For example, the specific transmission state may correspond to transmission performed by a transmitting UE on resources not included in available resources. For example, the specific transmission state may correspond to transmission performed by a transmitting UE on a resource not included in a candidate group for reservation

resources. For example, the specific transmission state may correspond to transmission including CSI reporting. For example, the specific transmission state may correspond to transmission including only CSI reporting. For example, a threshold value used in various embodiments of the present disclosure may be configured or pre-configure for a UE differently for each of the specific transmission states (e.g., initial transmission or retransmission).

For example, various embodiments of the present disclosure may be used when a UE selects a proposed resource to be instructed or transmitted to a third UE (e.g., expecting or suggesting that a third UE perform or avoid sidelink transmission to a UE for all or part of the corresponding resource). In the above application method, the process of comparing the threshold value with the new reference value in the embodiment of the present disclosure may be omitted, and the priority may be determined using the new reference value itself.

For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each resource pool. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each service type. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each (transmission and/or reception) priority. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each QoS parameter. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each power control mode. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each congestion control level. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each of whether HARQ feedback is. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each HARQ feedback option. For example, whether to apply various embodiments of the present disclosure may be configured or pre-configure for a UE for each cast type. For example, when the congestion level is high, the benefits of the additional resource exclusion process or the proposal process may be small.

FIG. 11 is a flowchart showing a method for a first device to perform sidelink communication, according to an embodiment of the present disclosure.

Operations disclosed in the flowchart of FIG. 11 may be performed in combination with various embodiments of the present disclosure. In one example, the operations disclosed in the flowchart of FIG. 11 may be performed based on at least one of the devices shown in FIGS. 13 to 18. In one example, the first device of FIG. 11 may correspond to the first wireless device 100 of FIG. 14 described below, and the second device may correspond to the second wireless device 200 of FIG. 14. In another example, the first device of FIG. 11 may correspond to the second wireless device 200 of FIG. 14 described later, and the second device may correspond to the first wireless device 100.

In step S1110, a first device according to an embodiment may reserve a first reservation resource, based on a first sensing operation within a sensing window.

In step S1120, a first device according to an embodiment may reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window.

In step S1130, a first device according to an embodiment may obtain a first reference signal received power (RSRP) value for the first reservation resource.

In step S1140, a first device according to an embodiment may obtain a second RSRP value for the second reservation resource.

In step S1150, a first device according to an embodiment may derive a sum value based on the first RSRP value and the second RSRP value.

In step S1160, a first device according to an embodiment may exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

In one embodiment, the sum value may be derived by summing the first RSRP value and the second RSRP value.

In one embodiment, the sum value may be an average value of the first RSRP value and the second RSRP value.

In one embodiment, the sum value may be a weighted average of the first RSRP value and the second RSRP value.

In one embodiment, the first threshold value may be configured on the first device, based on at least one of a resource pool, a first priority value for the first reservation resource, a second priority value for the second reservation resource, or a third priority value for data to be transmitted.

In one embodiment, the first threshold value may be configured of pre-configured on the first device to be different for each range of power values per physical resource block (PRB) related to sidelink transmission of the first device.

In one embodiment, whether to exclude the first reservation resource from the available resources for the sidelink communication of the first device may be determined based on a first ratio value which is ratio of a reference transmit power value and the first RSRP value. And, whether to exclude the second reservation resource from the available resources for the sidelink communication of the first device may be determined based on a second ratio value which is ratio of the reference transmit power value and the second RSRP value.

In one embodiment, the first reservation resource may be excluded from the available resources for the sidelink communication of the first device, based on the first ratio value being less than a second threshold value. And, the second reservation resource may be excluded from the available resources for the sidelink communication of the first device, based on the second ratio value being less than a third threshold value.

In one embodiment, whether to exclude the first reservation resource and the second reservation resource from the available resources for the sidelink communication of the first device may be determined based on a third ratio value which is ratio of a reference transmit power value and the sum value.

In one embodiment, the first reservation resource and the second reservation resource may be excluded from the available resources for the sidelink communication of the first device, based on the third ratio value being less than a fourth threshold value.

According to an embodiment of the present disclosure, a first device for performing sidelink communication may be proposed. The first device may comprise: at least one memory storing instructions; at least one transceiver; and at least one processor connected to the at least one memory and the at least one transceiver, wherein the at least one processor execute the instructions to: reserve a first reservation resource, based on a first sensing operation within a sensing

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window; reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window; obtain a first reference signal received power (RSRP) value for the first reservation resource; obtain a second RSRP value for the second reservation resource; derive a sum value based on the first RSRP value and the second RSRP value; and exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

According to an embodiment of the present disclosure, a device (or a chip(set)) adapted to control a first user equipment (UE) may be proposed. The device may comprise: one or more processors; and one or more memories operably connectable to the one or more processors and storing instructions, wherein the one or more processors execute the instructions to: reserve a first reservation resource, based on a first sensing operation within a sensing window; reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window; obtain a first reference signal received power (RSRP) value for the first reservation resource; obtain a second RSRP value for the second reservation resource; derive a sum value based on the first RSRP value and the second RSRP value; and exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first UE, based on the sum value exceeding a first threshold value.

In one example, the first UE in the above embodiment may represent the first device described throughout the present disclosure. In one example, the at least one processor, the at least one memory, etc. in the device for controlling the first UE may be implemented as separate sub chips, alternatively, at least two or more components may be implemented through one sub-chip.

According to an embodiment of the present disclosure, a non-transitory computer-readable storage medium storing instructions may be proposed. The instructions, when executed, may cause a first device to: reserve a first reservation resource, based on a first sensing operation within a sensing window; reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window; obtain a first reference signal received power (RSRP) value for the first reservation resource; obtain a second RSRP value for the second reservation resource; derive a sum value based on the first RSRP value and the second RSRP value; and exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

FIG. 12 is a flowchart showing a method for a second device to perform sidelink communication, according to an embodiment of the present disclosure.

Operations disclosed in the flowchart of FIG. 12 may be performed in combination with various embodiments of the present disclosure. In one example, the operations disclosed in the flowchart of FIG. 12 may be performed based on at least one of the devices shown in FIGS. 13 to 18. In one example, the second device of FIG. 12 may correspond to the second wireless device 200 of FIG. 14 described below, and the first device may correspond to the first wireless device 100 of FIG. 14. In another example, the second device of FIG. 12 may correspond to the first wireless device

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100 described later, and the first device may correspond to the second wireless device 200 of FIG. 14.

A first device according to an embodiment may reserve a first reservation resource, based on a first sensing operation within a sensing window.

A first device according to an embodiment may reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window.

A first device according to an embodiment may obtain a first reference signal received power (RSRP) value for the first reservation resource.

A first device according to an embodiment may obtain a second RSRP value for the second reservation resource.

A first device according to an embodiment may derive a sum value based on the first RSRP value and the second RSRP value.

A first device according to an embodiment may exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

In one embodiment, the sum value may be derived by summing the first RSRP value and the second RSRP value.

In one embodiment, the sum value may be an average value of the first RSRP value and the second RSRP value.

In one embodiment, the sum value may be a weighted average of the first RSRP value and the second RSRP value.

In one embodiment, the first threshold value may be configured on the first device, based on at least one of a resource pool, a first priority value for the first reservation resource, a second priority value for the second reservation resource, or a third priority value for data to be transmitted.

In one embodiment, the first threshold value may be configured of pre-configured on the first device to be different for each range of power values per physical resource block (PRB) related to sidelink transmission of the first device.

In one embodiment, whether to exclude the first reservation resource from the available resources for the sidelink communication of the first device may be determined based on a first ratio value which is ratio of a reference transmit power value and the first RSRP value. And, whether to exclude the second reservation resource from the available resources for the sidelink communication of the first device may be determined based on a second ratio value which is ratio of the reference transmit power value and the second RSRP value.

In one embodiment, the first reservation resource may be excluded from the available resources for the sidelink communication of the first device, based on the first ratio value being less than a second threshold value. And, the second reservation resource may be excluded from the available resources for the sidelink communication of the first device, based on the second ratio value being less than a third threshold value.

In one embodiment, whether to exclude the first reservation resource and the second reservation resource from the available resources for the sidelink communication of the first device may be determined based on a third ratio value which is ratio of a reference transmit power value and the sum value.

In one embodiment, the first reservation resource and the second reservation resource may be excluded from the available resources for the sidelink communication of the first device, based on the third ratio value being less than a fourth threshold value.

In step **S1210**, a second device according to an embodiment may receive, from the first device, a PSSCH through some resources among available resources for sidelink communication of the first device. Here, the available resources may be resources excluding reservation resources excluded based on the foregoing embodiment.

In step **S1220**, a second device according to an embodiment may receive, from the first device, a PSSCH related to a PSSCH, through some resources among available resources for sidelink communication of the first device.

According to an embodiment of the present disclosure, a second device for performing sidelink communication is proposed. The second device may comprise: at least one memory storing instructions; at least one transceiver; and at least one processor connected to the at least one memory and the at least one transceiver, wherein the at least one processor execute the instructions to: receive, from the first device, a PSSCH through some resources among available resources for sidelink communication of the first device. Here, the available resources may be resources excluding reservation resources excluded based on the foregoing embodiment; and receive, from the first device, a PSSCH related to a PSSCH, through some resources among available resources for sidelink communication of the first device.

Various embodiments of the present disclosure may be combined with each other.

Hereinafter, device(s) to which various embodiments of the present disclosure can be applied will be described.

The various descriptions, functions, procedures, proposals, methods, and/or operational flowcharts of the present disclosure described in this document may be applied to, without being limited to, a variety of fields requiring wireless communication/connection (e.g., 5G) between devices.

Hereinafter, a description will be given in more detail with reference to the drawings. In the following drawings/description, the same reference symbols may denote the same or corresponding hardware blocks, software blocks, or functional blocks unless described otherwise.

FIG. 13 shows a communication system **1**, in accordance with an embodiment of the present disclosure.

Referring to FIG. 13, a communication system **1** to which various embodiments of the present disclosure are applied includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot **100a**, vehicles **100b-1** and **100b-2**, an eXtended Reality (XR) device **100c**, a hand-held device **100d**, a home appliance **100e**, an Internet of Things (IoT) device **100f**, and an Artificial Intelligence (AI) device/server **400**. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home

appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be implemented as wireless devices and a specific wireless device **200a** may operate as a BS/network node with respect to other wireless devices.

Here, wireless communication technology implemented in wireless devices **100a** to **100f** of the present disclosure may include Narrowband Internet of Things for low-power communication in addition to LTE, NR, and 6G. In this case, for example, NB-IoT technology may be an example of Low Power Wide Area Network (LPWAN) technology and may be implemented as standards such as LTE Cat NB1, and/or LTE Cat NB2, and is not limited to the name described above. Additionally or alternatively, the wireless communication technology implemented in the wireless devices **100a** to **100f** of the present disclosure may perform communication based on LTE-M technology. In this case, as an example, the LTE-M technology may be an example of the LPWAN and may be called by various names including enhanced Machine Type Communication (eMTC), and the like. For example, the LTE-M technology may be implemented as at least any one of various standards such as 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-Bandwidth Limited (non-BL), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M, and is not limited to the name described above. Additionally or alternatively, the wireless communication technology implemented in the wireless devices **100a** to **100f** of the present disclosure may include at least one of Bluetooth, Low Power Wide Area Network (LPWAN), and ZigBee considering the low-power communication, and is not limited to the name described above. As an example, the ZigBee technology may generate personal area networks (PAN) related to small/low-power digital communication based on various standards including IEEE 802.15.4, and the like, and may be called by various names.

The wireless devices **100a** to **100f** may be connected to the network **300** via the BSs **200**. An AI technology may be applied to the wireless devices **100a** to **100f** and the wireless devices **100a** to **100f** may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices **100a** to **100f** may communicate with each other through the BSs **200**/network **300**, the wireless devices **100a** to **100f** may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles **100b-1** and **100b-2** may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100a** to **100f**.

Wireless communication/connections **150a**, **150b**, or **150c** may be established between the wireless devices **100a** to **100f**/BS **200**, or BS **200**/BS **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication **150a**, sidelink communication **150b** (or, D2D communication), or inter BS communication (e.g. relay, Integrated Access Backhaul (IAB)). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections **150a** and **150b**. For example, the wireless communication/connections **150a** and **150b** may transmit/receive signals through various physical channels. To

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this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

FIG. 14 shows wireless devices, in accordance with an embodiment of the present disclosure.

Referring to FIG. 14, a first wireless device 100 and a second wireless device 200 may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device 100 and the second wireless device 200} may correspond to {the wireless device 100x and the BS 200} and/or {the wireless device 100x and the wireless device 100x} of FIG. 13.

The first wireless device 100 may include one or more processors 102 and one or more memories 104 and additionally further include one or more transceivers 106 and/or one or more antennas 108. The processor(s) 102 may control the memory(s) 104 and/or the transceiver(s) 106 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 102 may process information within the memory(s) 104 to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) 106. The processor(s) 102 may receive radio signals including second information/signals through the transceiver 106 and then store information obtained by processing the second information/signals in the memory(s) 104. The memory(s) 104 may be connected to the processor(s) 102 and may store a variety of information related to operations of the processor(s) 102. For example, the memory(s) 104 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 102 or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 102 and the memory(s) 104 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 106 may be connected to the processor(s) 102 and transmit and/or receive radio signals through one or more antennas 108. Each of the transceiver(s) 106 may include a transmitter and/or a receiver. The transceiver(s) 106 may be interchangeably used with Radio Frequency (RF) unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

The second wireless device 200 may include one or more processors 202 and one or more memories 204 and additionally further include one or more transceivers 206 and/or one or more antennas 208. The processor(s) 202 may control the memory(s) 204 and/or the transceiver(s) 206 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 202 may process information within the memory(s) 204 to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) 206. The processor(s) 202 may receive radio signals including fourth information/signals through the transceiver(s) 106 and then store information obtained by processing the fourth information/signals in the memory(s) 204. The memory(s) 204 may be connected to the processor(s) 202 and may store a variety of information related to operations of the processor(s) 202. For example, the memo-

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ry(s) 204 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 202 or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 202 and the memory(s) 204 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 206 may be connected to the processor(s) 202 and transmit and/or receive radio signals through one or more antennas 208. Each of the transceiver(s) 206 may include a transmitter and/or a receiver. The transceiver(s) 206 may be interchangeably used with RF unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices 100 and 200 will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors 102 and 202. For example, the one or more processors 102 and 202 may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors 102 and 202 may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors 102 and 202 may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors 102 and 202 may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers 106 and 206. The one or more processors 102 and 202 may receive the signals (e.g., baseband signals) from the one or more transceivers 106 and 206 and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

The one or more processors 102 and 202 may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors 102 and 202 may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors 102 and 202. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors 102 and 202 or stored in the one or more memories 104 and 204 so as to be driven by the one or more processors 102 and 202. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this

document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

FIG. **15** shows a signal process circuit for a transmission signal, in accordance with an embodiment of the present disclosure.

Referring to FIG. **15**, a signal processing circuit **1000** may include scramblers **1010**, modulators **1020**, a layer mapper **1030**, a precoder **1040**, resource mappers **1050**, and signal generators **1060**. An operation/function of FIG. **15** may be performed, without being limited to, the processors **102** and **202** and/or the transceivers **106** and **206** of FIG. **14**. Hardware elements of FIG. **15** may be implemented by the processors **102** and **202** and/or the transceivers **106** and **206**

of FIG. **14**. For example, blocks **1010** to **1060** may be implemented by the processors **102** and **202** of FIG. **14**. Alternatively, the blocks **1010** to **1050** may be implemented by the processors **102** and **202** of FIG. **14** and the block **1060** may be implemented by the transceivers **106** and **206** of FIG. **14**.

Codewords may be converted into radio signals via the signal processing circuit **1000** of FIG. **15**. Herein, the codewords are encoded bit sequences of information blocks. The information blocks may include transport blocks (e.g., a UL-SCH transport block, a DL-SCH transport block). The radio signals may be transmitted through various physical channels (e.g., a PUSCH and a PDSCH).

Specifically, the codewords may be converted into scrambled bit sequences by the scramblers **1010**. Scramble sequences used for scrambling may be generated based on an initialization value, and the initialization value may include ID information of a wireless device. The scrambled bit sequences may be modulated to modulation symbol sequences by the modulators **1020**. A modulation scheme may include pi/2-Binary Phase Shift Keying (pi/2-BPSK), m-Phase Shift Keying (m-PSK), and m-Quadrature Amplitude Modulation (m-QAM). Complex modulation symbol sequences may be mapped to one or more transport layers by the layer mapper **1030**. Modulation symbols of each transport layer may be mapped (precoded) to corresponding antenna port(s) by the precoder **1040**. Outputs z of the precoder **1040** may be obtained by multiplying outputs y of the layer mapper **1030** by an $N \times M$ precoding matrix W . Herein, N is the number of antenna ports and M is the number of transport layers. The precoder **1040** may perform precoding after performing transform precoding (e.g., DFT) for complex modulation symbols. Alternatively, the precoder **1040** may perform precoding without performing transform precoding.

The resource mappers **1050** may map modulation symbols of each antenna port to time-frequency resources. The time-frequency resources may include a plurality of symbols (e.g., a CP-OFDMA symbols and DFT-s-OFDMA symbols) in the time domain and a plurality of subcarriers in the frequency domain. The signal generators **1060** may generate radio signals from the mapped modulation symbols and the generated radio signals may be transmitted to other devices through each antenna. For this purpose, the signal generators **1060** may include Inverse Fast Fourier Transform (IFFT) modules, Cyclic Prefix (CP) inserters, Digital-to-Analog Converters (DACs), and frequency up-converters.

Signal processing procedures for a signal received in the wireless device may be configured in a reverse manner of the signal processing procedures **1010** to **1060** of FIG. **15**. For example, the wireless devices (e.g., **100** and **200** of FIG. **14**) may receive radio signals from the exterior through the antenna ports/transceivers. The received radio signals may be converted into baseband signals through signal restorers. To this end, the signal restorers may include frequency downlink converters, Analog-to-Digital Converters (ADCs), CP remover, and Fast Fourier Transform (FFT) modules. Next, the baseband signals may be restored to codewords through a resource demapping procedure, a postcoding procedure, a demodulation processor, and a descrambling procedure. The codewords may be restored to original information blocks through decoding. Therefore, a signal processing circuit (not illustrated) for a reception signal may include signal restorers, resource demappers, a postcoder, demodulators, descramblers, and decoders.

FIG. **16** shows another example of a wireless device, in accordance with an embodiment of the present disclosure.

The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. 13).

Referring to FIG. 16, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 14 and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204** of FIG. 14. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. 14. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. 13), the vehicles (**100b-1** and **100b-2** of FIG. 13), the XR device (**100c** of FIG. 13), the hand-held device (**100d** of FIG. 13), the home appliance (**100e** of FIG. 13), the IoT device (**100f** of FIG. 13), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. 13), the BSs (**200** of FIG. 13), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

In FIG. 16, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a Random Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

Hereinafter, an example of implementing FIG. 16 will be described in detail with reference to the drawings.

FIG. 17 shows a hand-held device, in accordance with an embodiment of the present disclosure. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), or a portable computer (e.g., a notebook). The hand-held device may be referred to as a mobile station (MS), a user terminal (UT), a Mobile Subscriber Station (MSS), a Subscriber Station (SS), an Advanced Mobile Station (AMS), or a Wireless Terminal (WT).

Referring to FIG. 17, a hand-held device **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a memory unit **130**, a power supply unit **140a**, an interface unit **140b**, and an I/O unit **140c**. The antenna unit **108** may be configured as a part of the communication unit **110**. Blocks **110** to **130/140a** to **140c** correspond to the blocks **110** to **130/140** of FIG. 16, respectively.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit **120** may perform various operations by controlling constituent elements of the hand-held device **100**. The control unit **120** may include an Application Processor (AP). The memory unit **130** may store data/parameters/programs/code/commands needed to drive the hand-held device **100**. The memory unit **130** may store input/output data/information. The power supply unit **140a** may supply power to the hand-held device **100** and include a wired/wireless charging circuit, a battery, etc. The interface unit **140b** may support connection of the hand-held device **100** to other external devices. The interface unit **140b** may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit **140c** may input or output video information/signals, audio information/signals, data, and/or information input by a user. The I/O unit **140c** may include a camera, a microphone, a user input unit, a display unit **140d**, a speaker, and/or a haptic module.

As an example, in the case of data communication, the I/O unit **140c** may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit **130**. The communication unit **110** may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. The communication unit **110** may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit **130** and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit **140c**.

FIG. 18 shows a vehicle or an autonomous vehicle, in accordance with an embodiment of the present disclosure. The vehicle or autonomous vehicle may be implemented by a mobile robot, a car, a train, a manned/unmanned Aerial Vehicle (AV), a ship, etc.

Referring to FIG. 18, a vehicle or autonomous vehicle **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a driving unit **140a**, a power supply unit **140b**, a sensor unit **140c**, and an autonomous driving unit **140d**. The antenna unit **108** may be configured as a part of the communication unit **110**. The blocks **110/130/140a** to **140d** correspond to the blocks **110/130/140** of FIG. 16, respectively.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external

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devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit **120** may perform various operations by controlling elements of the vehicle or the autonomous vehicle **100**. The control unit **120** may include an Electronic Control Unit (ECU). The driving unit **140a** may cause the vehicle or the autonomous vehicle **100** to drive on a road. The driving unit **140a** may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit **140b** may supply power to the vehicle or the autonomous vehicle **100** and include a wired/wireless charging circuit, a battery, etc. The sensor unit **140c** may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit **140c** may include an Inertial Measurement Unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit **140d** may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

For example, the communication unit **110** may receive map data, traffic information data, etc. from an external server. The autonomous driving unit **140d** may generate an autonomous driving path and a driving plan from the obtained data. The control unit **120** may control the driving unit **140a** such that the vehicle or the autonomous vehicle **100** may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit **110** may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit **140c** may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit **140d** may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit **110** may transfer information about a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous vehicles and provide the predicted traffic information data to the vehicles or the autonomous vehicles.

Claims in the present description can be combined in a various way. For instance, technical features in method claims of the present description can be combined to be implemented or performed in an apparatus, and technical features in apparatus claims can be combined to be implemented or performed in a method. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in an apparatus. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in a method.

What is claimed is:

1. A method for performing, by a first device, sidelink communication, the method comprising:

reserving a first reservation resource, based on a first sensing operation within a sensing window;

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reserving a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window; obtaining a first reference signal received power (RSRP) value for the first reservation resource;

obtaining a second RSRP value for the second reservation resource;

deriving a sum value based on the first RSRP value and the second RSRP value; and

excluding the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

2. The method of claim 1, wherein the sum value is derived by summing the first RSRP value and the second RSRP value.

3. The method of claim 1, wherein the sum value is an average value of the first RSRP value and the second RSRP value.

4. The method of claim 1, wherein the sum value is a weighted average of the first RSRP value and the second RSRP value.

5. The method of claim 1, wherein the first threshold value is configured on the first device, based on at least one of a resource pool, a first priority value for the first reservation resource, a second priority value for the second reservation resource, or a third priority value for data to be transmitted.

6. The method of claim 1, wherein the first threshold value is configured of pre-configured on the first device to be different for each range of power values per physical resource block (PRB) related to sidelink transmission of the first device.

7. The method of claim 1, wherein whether to exclude the first reservation resource from the available resources for the sidelink communication of the first device is determined based on a first ratio value which is ratio of a reference transmit power value and the first RSRP value, and wherein whether to exclude the second reservation resource from the available resources for the sidelink communication of the first device is determined based on a second ratio value which is ratio of the reference transmit power value and the second RSRP value.

8. The method of claim 7, wherein the first reservation resource is excluded from the available resources for the sidelink communication of the first device, based on the first ratio value being less than a second threshold value, and wherein the second reservation resource is excluded from the available resources for the sidelink communication of the first device, based on the second ratio value being less than a third threshold value.

9. The method of claim 1, wherein whether to exclude the first reservation resource and the second reservation resource from the available resources for the sidelink communication of the first device is determined based on a third ratio value which is ratio of a reference transmit power value and the sum value.

10. The method of claim 9, wherein the first reservation resource and the second reservation resource are excluded from the available resources for the sidelink communication of the first device, based on the third ratio value being less than a fourth threshold value.

11. A first device for performing sidelink communication, the first device comprising:

at least one memory storing instructions;

at least one transceiver; and

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at least one processor connected to the at least one memory and the at least one transceiver, wherein the at least one processor execute the instructions to:

reserve a first reservation resource, based on a first sensing operation within a sensing window;

reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window;

obtain a first reference signal received power (RSRP) value for the first reservation resource;

obtain a second RSRP value for the second reservation resource;

derive a sum value based on the first RSRP value and the second RSRP value; and

exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first device, based on the sum value exceeding a first threshold value.

12. A device adapted to control a first user equipment (UE), the device comprising:

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one or more processors; and

one or more memories operably connectable to the one or more processors and storing instructions, wherein the one or more processors execute the instructions to:

reserve a first reservation resource, based on a first sensing operation within a sensing window;

reserve a second reservation resource partially overlapping with the first reservation resource, based on a second sensing operation within the sensing window;

obtain a first reference signal received power (RSRP) value for the first reservation resource;

obtain a second RSRP value for the second reservation resource;

derive a sum value based on the first RSRP value and the second RSRP value; and

exclude the first reservation resource and the second reservation resource from available resources for sidelink communication of the first UE, based on the sum value exceeding a first threshold value.

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